

Convexity & Concavity

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Section 1

Convex sets

Convex combination: definition

Let $m \geq 1$ and let $x_1, \dots, x_m \in \mathbb{R}^n$.

Convex combination. A point $z \in \mathbb{R}^n$ is a convex combination of $x_1, \dots, x_m$ if there are weights $\lambda_1, \dots, \lambda_m \in \mathbb{R}$ such that

$$z = \sum_{i=1}^m \lambda_i x_i, \quad \lambda_i \geq 0 \text{ for every } i, \quad \sum_{i=1}^m \lambda_i = 1.$$

Video: Dr. Will Wood, *Convex sets: introduction, definition and examples*.

NONNEGATIVE WEIGHTS

Every point enters with a weight that is at least zero.

WEIGHTS SUM TO ONE

The coefficients form a weighted average of the points.

Linear, affine, conic, and convex combinations

Start from the coefficients, then impose restrictions

For $x_1, \dots, x_m \in \mathbb{R}^n$, start from $z = \sum_{i=1}^m \lambda_i x_i$ with $\lambda_i \in \mathbb{R}$, then classify the combination by the coefficient conditions.

Combination	Coefficient condition	Sum condition
Linear	$\lambda_i \in \mathbb{R}$	none
Affine	$\lambda_i \in \mathbb{R}$	$\sum_{i=1}^m \lambda_i = 1$
Conic	$\lambda_i \geq 0$ for every i	none
Convex	$\lambda_i \geq 0$ for every i	$\sum_{i=1}^m \lambda_i = 1$

A convex combination satisfies $\sum_i \lambda_i = 1$, as an affine combination does, and $\lambda_i \geq 0$, as a conic combination does. For two distinct points x, y , $\lambda \in \mathbb{R}$ traces the full line through them, while $\lambda \in [0, 1]$ traces the joining segment.

Cones and convex cones

Closed under nonnegative scaling, then also under addition

A set $C \subseteq \mathbb{R}^n$ is a **cone** if

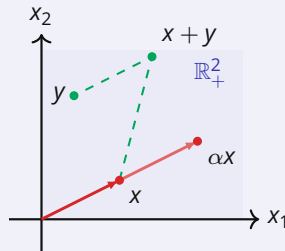
$$x \in C, \alpha \geq 0 \implies \alpha x \in C,$$

and a **convex cone** if

$$x, y \in C, \alpha, \beta \geq 0 \implies \alpha x + \beta y \in C.$$

$\beta = 0$ gives the cone property and $\alpha + \beta = 1$ gives convexity; conversely, in a cone that is convex, $\alpha + \beta = 0$ forces $\alpha = \beta = 0$ and $\alpha x + \beta y = 0 \in C$, while for $\alpha + \beta > 0$ the point $(\alpha + \beta) \left[\frac{\alpha}{\alpha + \beta} x + \frac{\beta}{\alpha + \beta} y \right]$ lies in C . So a convex cone contains every conic combination of its points. $\mathbb{R}_+^2$ is a convex cone: the coordinates of $\alpha x + \beta y$ are sums of nonnegative numbers.

Further reading: [Boyd and Vandenberghe, *Convex Optimization*, §2.1.5.](#)



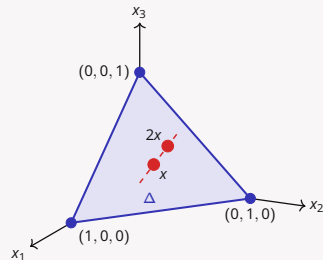
The simplex is convex but is not a cone

A convex set that fails the scaling test

Let $\Delta = \{x \in \mathbb{R}_+^n : \sum_{i=1}^n x_i = 1\}$. For $x, y \in \Delta$ and $\lambda \in [0, 1]$, the point $\lambda x + (1 - \lambda)y$ has nonnegative coordinates and

$$\sum_i (\lambda x_i + (1 - \lambda)y_i) = \lambda \cdot 1 + (1 - \lambda) \cdot 1 = 1,$$

so Δ is convex. For any $x \in \Delta$, the coordinates of $2x$ sum to 2, so $2x \notin \Delta$ and Δ is not a cone. For $n = 3$, Δ is the triangle with vertices $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$; its center $x = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ lies in Δ , while $2x = (\frac{2}{3}, \frac{2}{3}, \frac{2}{3})$ on the same ray from the origin lies above it.



A convex combination in $\mathbb{R}^2$

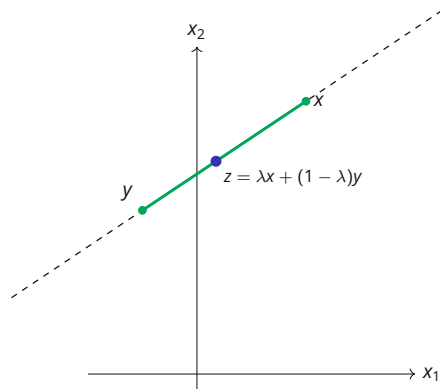


Figure 1 · A segment in $\mathbb{R}^2$.

A convex combination of x and y is any point on the segment joining them.

Convex combinations

Computing a convex combination

POINTS

$$x_1 = (0, 0), \quad x_2 = (2, 0), \quad x_3 = (0, 2).$$

WEIGHTS

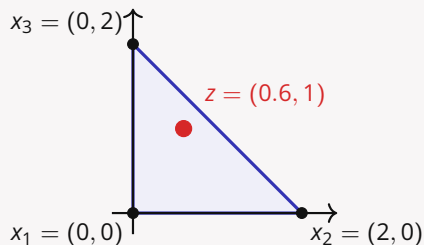
$$(\lambda_1, \lambda_2, \lambda_3) = (0.2, 0.3, 0.5).$$

All three weights are nonnegative, and

$$\lambda_1 + \lambda_2 + \lambda_3 = 1.$$

CALCULATION

$$\begin{aligned} z &= 0.2x_1 + 0.3x_2 + 0.5x_3 \\ &= (0.6, 1). \end{aligned}$$



The shaded triangle is the set of all convex combinations of x_1, x_2, x_3 . The red circle marks a particular point z from them.

Testing membership with a linear system

Solve for the weights, then check their signs

For $x_1 = (0, 0)$, $x_2 = (2, 0)$, and $x_3 = (0, 2)$, a target $w = (w_1, w_2)$ is a convex combination of them when the system

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{bmatrix} = \begin{bmatrix} 1 \\ w_1 \\ w_2 \end{bmatrix}$$

has a solution with $\lambda_1, \lambda_2, \lambda_3 \geq 0$. For $w = (0.6, 1)$, the solution is

$$(\lambda_1, \lambda_2, \lambda_3) = (0.2, 0.3, 0.5),$$

so w belongs to the shaded triangle on the preceding page.

Definition

Convex set. $S \subseteq \mathbb{R}^n$ is convex if

$$\forall x, y \in S, \forall \lambda \in [0, 1] : \quad \lambda x + (1 - \lambda)y \in S.$$

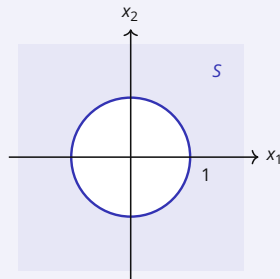
- ◆ Equivalently, S contains the entire segment between **any** two of its points.
- ◆ Visually: every line segment joining two points of S stays inside S .
- ◆ Induction on the number of points extends the two-point condition to every convex combination of finitely many points of S (exercise below).

Negating the definition of convexity

Write the negation of the definition of a convex set.
Then use the negated statement to show that

$$S = \{(x_1, x_2) \in \mathbb{R}^2 : x_1^2 + x_2^2 \geq 1\},$$

the shaded region on and outside the unit circle, is not convex.



Negating the definition of convexity

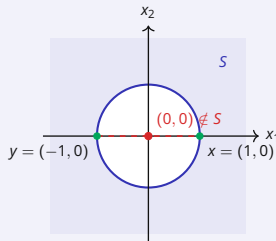
A set S is not convex if

$$\exists x, y \in S, \exists \lambda \in [0, 1] \text{ such that } \lambda x + (1 - \lambda)y \notin S.$$

Take

$$x = (1, 0), \quad y = (-1, 0), \quad \lambda = \frac{1}{2}.$$

Both endpoints belong to S , while their midpoint is $(0, 0)$, and $0^2 + 0^2 = 0 < 1$. Thus the midpoint is outside S , so S is not convex: the segment joining x and y crosses the hole.



From two points to finitely many

Let $C \subseteq \mathbb{R}^n$ be convex. Show that for every $m \geq 1$, all $x_1, \dots, x_m \in C$, and all weights $\lambda_i \geq 0$ with $\sum_{i=1}^m \lambda_i = 1$,

$$\sum_{i=1}^m \lambda_i x_i \in C.$$

Hint: induction on m .

From two points to finitely many

For $m = 1$ the weight is $\lambda_1 = 1$ and the combination is x_1 ; for $m = 2$ the claim is the definition. Let $m \geq 3$ and assume the claim for $m - 1$ points. If $\lambda_m = 1$, the combination equals x_m . Otherwise $\mu = 1 - \lambda_m = \sum_{i < m} \lambda_i > 0$, and the weights λ_i/μ ($i < m$) are nonnegative and sum to one, so

$$z = \sum_{i < m} \frac{\lambda_i}{\mu} x_i \in C$$

by the induction hypothesis. Finally

$$\sum_{i=1}^m \lambda_i x_i = \mu z + \lambda_m x_m, \quad \mu + \lambda_m = 1, \quad \mu, \lambda_m \geq 0,$$

which lies in C by the two-point condition.

Extreme points cannot be decomposed inside the set

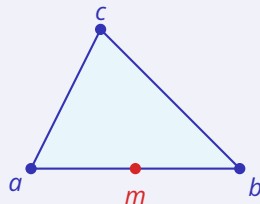
Extreme points

A point $x \in C$ is **extreme** if

$$x = \lambda y + (1 - \lambda)z, \quad y, z \in C, \quad \lambda \in (0, 1)$$

implies $y = z = x$.

For noncollinear a, b, c , the three vertices of $C = \text{conv}\{a, b, c\}$ are extreme. The midpoint $m = \frac{1}{2}a + \frac{1}{2}b$ is not.



Convex hull: generating a convex set from a non-convex set

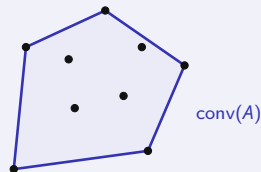
Convex hull of a set

For $A \subseteq \mathbb{R}^n$,

$$\text{conv}(A) = \left\{ \sum_{i=1}^m \lambda_i x_i : m \geq 1, x_i \in A, \lambda_i \geq 0, \sum_{i=1}^m \lambda_i = 1 \right\}.$$

Equivalently, $\text{conv}(A)$ is the smallest convex set containing A .

Further reading: [Boyd and Vandenberghe, convex-set slides](#).



A finite set A (dots) and the shaded $\text{conv}(A)$.

How many points a convex combination needs

Carathéodory's theorem and the extreme points of a compact convex set

Carathéodory. If $A \subseteq \mathbb{R}^n$, every $x \in \text{conv}(A)$ is a convex combination of at most $n + 1$ points of A .

Write $x = \sum_{i=1}^m \lambda_i x_i$ with all $\lambda_i > 0$ and $m \geq n + 2$. The $m - 1 \geq n + 1$ vectors $x_i - x_m$ are linearly dependent, so there are $\mu_1, \dots, \mu_m$, with $\sum_i \mu_i = 0$ and some $\mu_i > 0$, such that $\sum_i \mu_i x_i = 0$. Then $x = \sum_i (\lambda_i - t\mu_i)x_i$ for every t , and the new weights still sum to one. Increase t from 0 until the first weight reaches zero, at $t^* = \min\{\lambda_i/\mu_i : \mu_i > 0\}$: all weights stay nonnegative and at least one vanishes, so x uses $m - 1$ points. Repeat until $m \leq n + 1$.

Extreme points. A compact convex set $C \subseteq \mathbb{R}^n$ is the convex hull of its extreme points (Minkowski; the Krein–Milman theorem in finite dimension).

Further reading: Bertsekas, *Convex Optimization Theory*, Prop. 1.2.1 and Exercise 2.2; MIT 6.253, Lecture 3.

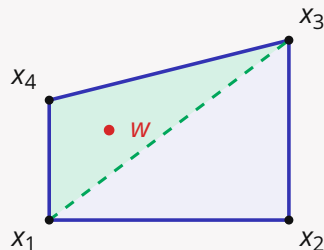
Three of four vertices suffice in the plane

Carathéodory in $\mathbb{R}^2$

Let $x_1 = (0, 0)$, $x_2 = (4, 0)$, $x_3 = (4, 3)$, $x_4 = (0, 2)$ and $w = (1, 1.5) \in \text{conv}\{x_1, x_2, x_3, x_4\}$. By Carathéodory, w is a convex combination of at most three vertices. The diagonal x_1x_3 splits the quadrilateral into two triangles, and w lies in $x_1x_3x_4$. Solving

$$\lambda_1 x_1 + \lambda_3 x_3 + \lambda_4 x_4 = w, \quad \lambda_1 + \lambda_3 + \lambda_4 = 1$$

as on the membership page gives $(\lambda_1, \lambda_3, \lambda_4) = (\frac{3}{8}, \frac{1}{4}, \frac{3}{8})$. With all four vertices the weights are no longer unique: also $w = 0.225x_1 + 0.1x_2 + 0.15x_3 + 0.525x_4$.



Budget set

Convexity of a budget set

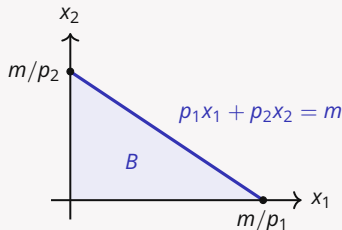
Let $p_1, p_2, m > 0$ and

$$B = \{x \in \mathbb{R}_+^2 : p_1 x_1 + p_2 x_2 \leq m\}.$$

Take any $x, y \in B$ and $\lambda \in [0, 1]$. Since $x, y \geq 0$, $\lambda x + (1 - \lambda)y \geq 0$. Also,

$$\begin{aligned} p \cdot (\lambda x + (1 - \lambda)y) &= \lambda p \cdot x + (1 - \lambda)p \cdot y \\ &\leq \lambda m + (1 - \lambda)m = m. \end{aligned}$$

Therefore $\lambda x + (1 - \lambda)y \in B$, so B is convex.



The budget line meets the axes at m/p_1 and m/p_2 ; B is the shaded triangle.

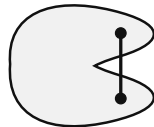
Convex & non-convex sets



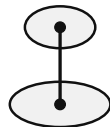
(a) convex



(b) convex



(c) not convex



(d) not convex

Basic properties

- ◆ **Arbitrary intersections** of convex sets are convex.
- ◆ **Linear combinations of convex sets are convex:** if S, T are convex and $\alpha, \beta \in \mathbb{R}$, then $\alpha S + \beta T = \{\alpha x + \beta y : x \in S, y \in T\}$ is convex.
- ◆ **Unions** of convex sets **need not** be convex.

Intersections

Let

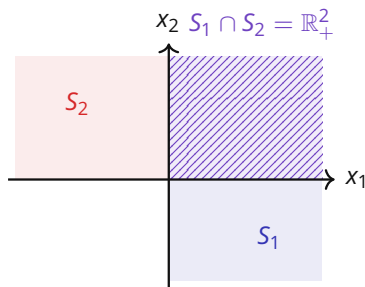
$$S_1 = \{x \in \mathbb{R}^2 : x_1 \geq 0\},$$

$$S_2 = \{x \in \mathbb{R}^2 : x_2 \geq 0\}.$$

A point belongs to $S_1 \cap S_2$ precisely when both coordinates are nonnegative. Therefore

$$S_1 \cap S_2 = \mathbb{R}_+^2,$$

which is convex.



The hatched first quadrant marks the points satisfying both inequalities.

Video: [Stanford EE364A, Lecture 2: convex sets and operations that preserve convexity.](#)

Intersections of convex sets

Let $\{S_i : i \in I\}$ be a collection of convex subsets of $\mathbb{R}^n$. Prove directly from the definition that

$$\bigcap_{i \in I} S_i$$

is convex.

Intersections of convex sets

If $x, y \in \bigcap_{i \in I} S_i$ and $t \in [0, 1]$, then $x, y \in S_i$ for every $i \in I$. Convexity of each S_i gives

$$tx + (1 - t)y \in S_i \quad \text{for every } i \in I.$$

Hence

$$tx + (1 - t)y \in \bigcap_{i \in I} S_i,$$

so the intersection is convex.

A linear system and its solution set

Let

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad S = \{x \in \mathbb{R}^3 : Ax = b\}.$$

Solve $Ax = b$, describe S , and prove directly that S is convex. Then, for matrices $A \in \mathbb{R}^{m \times n}$, $C \in \mathbb{R}^{k \times n}$ and vectors $b \in \mathbb{R}^m$, $d \in \mathbb{R}^k$, prove that

$$P = \{x \in \mathbb{R}^n : Ax \leq b, Cx = d\}$$

is convex, where $Ax \leq b$ holds coordinate by coordinate.

A linear system and its solution set

The equations are $x_1 + x_2 = 1$ and $x_2 + x_3 = 1$. Setting $x_2 = t$ gives

$$S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + t \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix} : t \in \mathbb{R} \right\} = x^0 + \ker A,$$

a line through the particular solution $x^0 = (1, 0, 1)$ in the direction of the null space of A . Since $A0 = 0 \neq b$, $0 \notin S$: the solution set is a subspace only when $b = 0$. If $x, y \in S$ and $\lambda \in [0, 1]$, then

$$A(\lambda x + (1 - \lambda)y) = \lambda Ax + (1 - \lambda)Ay = \lambda b + (1 - \lambda)b = b.$$

Hence $\lambda x + (1 - \lambda)y \in S$, so S is convex. The computation never used $\lambda \in [0, 1]$: for every real λ the point stays in S , so S contains the whole line through any two of its points (an affine set, two pages ahead). If the system has no solution, $S = \emptyset$, which is convex since there is no pair x, y to test.

A linear system and its solution set

For $P = \{x : Ax \leq b, Cx = d\}$, take $x, y \in P$ and $\lambda \in [0, 1]$. Multiplying a coordinatewise inequality by $\lambda \geq 0$ or by $1 - \lambda \geq 0$ preserves it, so

$$A(\lambda x + (1 - \lambda)y) = \lambda Ax + (1 - \lambda)Ay \leq \lambda b + (1 - \lambda)b = b,$$

and the equality rows give

$$C(\lambda x + (1 - \lambda)y) = \lambda d + (1 - \lambda)d = d.$$

Hence $\lambda x + (1 - \lambda)y \in P$, so P is convex. The budget set and the set T of the resource exercise are instances with one inequality row each.

Affine sets, hyperplanes, and halfspaces

Whole lines, one plane, one side of a plane

A set $S \subseteq \mathbb{R}^n$ is **affine** if

$$x, y \in S, \lambda \in \mathbb{R} \implies \lambda x + (1 - \lambda)y \in S :$$

it contains the whole line through any two of its points. Convexity asks the same only for $\lambda \in [0, 1]$, so every affine set is convex.

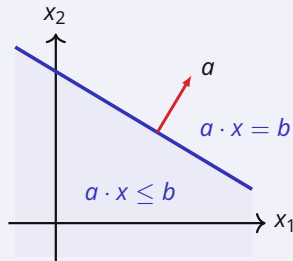
For $a \neq 0$ and $b \in \mathbb{R}$, the **hyperplane** $H = \{x : a \cdot x = b\}$ is affine:

$a \cdot (\lambda x + (1 - \lambda)y) = \lambda b + (1 - \lambda)b = b$ for every real λ . The

halfspace $\{x : a \cdot x \leq b\}$ is convex by the same computation with $\lambda \in [0, 1]$, which keeps the inequality.

The solution set $\{x : Ax = b\}$ of the preceding exercise is an intersection of hyperplanes, hence affine; $\{x : Ax \leq b, Cx = d\}$ is an intersection of halfspaces and hyperplanes, hence convex.

Further reading: [Boyd and Vandenberghe, §§2.1.2, 2.2.1, 2.2.4.](#)



Affine images and preimages

Convexity survives an affine map in both directions

Let $T(x) = Ax + b$ with $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$.

- ◆ If $S \subseteq \mathbb{R}^n$ is convex, its image $T(S) = \{Ax + b : x \in S\}$ is convex.
- ◆ If $D \subseteq \mathbb{R}^m$ is convex, its preimage $T^{-1}(D) = \{x \in \mathbb{R}^n : Ax + b \in D\}$ is convex.

Both follow from one identity, valid for all x, y and $\lambda \in [0, 1]$:

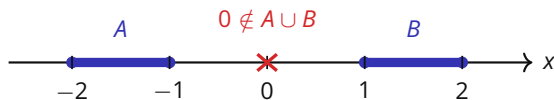
$$\lambda(Ax + b) + (1 - \lambda)(Ay + b) = A(\lambda x + (1 - \lambda)y) + b.$$

If $x, y \in S$, the right side is T of a point of S . If $Ax + b, Ay + b \in D$, the left side lies in D , so $\lambda x + (1 - \lambda)y \in T^{-1}(D)$.

The basic property $\alpha S + \beta T$ convex is the image of the convex set $S \times T$ under $(x, y) \mapsto \alpha x + \beta y$; the polyhedron $\{x : Ax \leq b\}$ is the preimage of $\mathbb{R}_-^m$ under $x \mapsto Ax - b$.

Further reading: [Boyd and Vandenberghe, §2.3.2](#).

Unions need not be convex



Take two convex intervals:

$$A = [-2, -1], \quad B = [1, 2].$$

Their union contains -1 and 1 , but it does not contain their midpoint:

$$-1, 1 \in A \cup B, \quad \frac{-1 + 1}{2} = 0 \notin A \cup B.$$

Hence $A \cup B$ is not convex.

A resource set and two subsets

Let

$$T = \{(x, y) \in \mathbb{R}^2 : 2x + y \leq 6\}, \quad C = \{(0, 0), (3, 0)\} \subset T.$$

- (a) Prove directly from the definition that T is convex.
- (b) Determine whether $T \cap C$ and $T \cup C$ are convex. Justify both answers.
- (c) Give a convex subset $D \subset T$, with at least two points, for which both $T \cap D$ and $T \cup D$ are convex.

A resource set and two subsets

If $u, v \in T$ and $t \in [0, 1]$, then

$$2(tu_1 + (1-t)v_1) + tu_2 + (1-t)v_2 = t(2u_1 + u_2) + (1-t)(2v_1 + v_2) \leq 6.$$

Thus $tu + (1-t)v \in T$, so T is convex. Since $C \subset T$,

$$T \cap C = C, \quad T \cup C = T.$$

The set C is not convex because $(3/2, 0) \notin C$, although it is the midpoint of $(0, 0)$ and $(3, 0)$. Hence $T \cap C$ is not convex, while $T \cup C$ is convex. One valid choice in part (c) is $D = \{(x, 0) : 0 \leq x \leq 3\}$: then $T \cap D = D$ and $T \cup D = T$.

Production sets and the profit function

Production plans and profit

A production set $Y \subseteq \mathbb{R}^n$ contains feasible netput vectors, with positive coordinates for outputs and negative coordinates for inputs. If Y is convex, mixtures of feasible production plans remain feasible.

$$\pi(p) = \sup_{y \in Y} p \cdot y, \quad p \in \mathbb{R}_+^n.$$

As a supremum of linear functions of p , π is convex wherever it is finite. Convexity of Y instead concerns feasible mixtures.

Profit function

Support function

Let

$$Y = \text{conv} \left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \begin{bmatrix} -2 \\ 3 \end{bmatrix} \right\}, \quad p = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

The linear payoff at the three vertices is

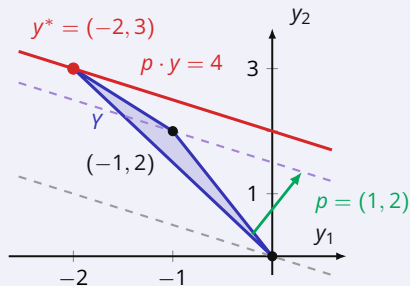
$$p \cdot \begin{bmatrix} 0 \\ 0 \end{bmatrix} = 0, \quad p \cdot \begin{bmatrix} -1 \\ 2 \end{bmatrix} = 3, \quad p \cdot \begin{bmatrix} -2 \\ 3 \end{bmatrix} = 4.$$

If $y = \sum_i \lambda_i y_i \in Y$, then linearity gives

$$p \cdot y = \sum_i \lambda_i (p \cdot y_i) \leq \max_i p \cdot y_i = 4.$$

Hence

$$\pi(p) = \sup_{y \in Y} p \cdot y = 4, \quad y^* = \begin{bmatrix} -2 \\ 3 \end{bmatrix}.$$



The level line $p \cdot y = 4$ supports Y at y^* .

Disjoint compact and closed convex sets can be strictly separated

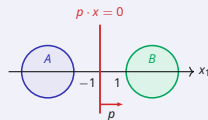
Separating hyperplanes

If $A \subset \mathbb{R}^n$ is nonempty, compact, and convex, and $B \subset \mathbb{R}^n$ is nonempty, closed, and convex, with $A \cap B = \emptyset$, then there exist $p \neq 0$ and $\alpha < \beta$ such that

$$p \cdot a \leq \alpha < \beta \leq p \cdot b \quad \forall a \in A, \forall b \in B.$$

Every level set $\{x : p \cdot x = t\}$ with $t \in (\alpha, \beta)$ is a separating hyperplane. The normal p gives the direction in which the two sets are ordered.

For the discs centered at $(-2, 0)$ and $(2, 0)$, both of radius one, $p = (1, 0)$ gives $\sup_{a \in A} p \cdot a = -1 < 1 = \inf_{b \in B} p \cdot b$, so $p \cdot x = 0$ separates them.



Video: [Constantine Caramanis, Separating hyperplanes.](#)

Every boundary point of a closed convex set has a support

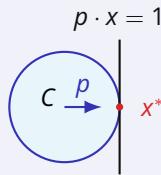
Supporting hyperplanes

Let $C \subset \mathbb{R}^n$ be nonempty, closed, and convex, and let $x^* \in \partial C$. There exists $p \neq 0$ such that

$$p \cdot x \leq p \cdot x^* \quad \forall x \in C.$$

The hyperplane $p \cdot x = p \cdot x^*$ supports C at x^* : it contains the boundary point and keeps the entire convex set in one closed half-space.

For the unit disc, $x^* = (1, 0)$ and $p = (1, 0)$ give the supporting line $p \cdot x = 1$.



Section 2

Convex & concave functions

Definitions via convex combinations

Let $S \subseteq \mathbb{R}^n$ be convex and $f : S \rightarrow \mathbb{R}$. Then the function f is

Convex. $f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$ for all $x, y \in S$ and $\lambda \in [0, 1]$.

Concave. $f(\lambda x + (1 - \lambda)y) \geq \lambda f(x) + (1 - \lambda)f(y)$ for all $x, y \in S$ and $\lambda \in [0, 1]$.

Strict convexity or concavity replaces the weak inequality by a strict one when $x \neq y$ and $\lambda \in (0, 1)$.

Here, convex and concave functions are defined only on convex domains. (Things are different in discrete convex analysis, which develops alternative notions of convexity for discrete domains.)

Video: [Arizona Math Camp, Concave and convex functions.](#)

Chord above, chord on, chord below

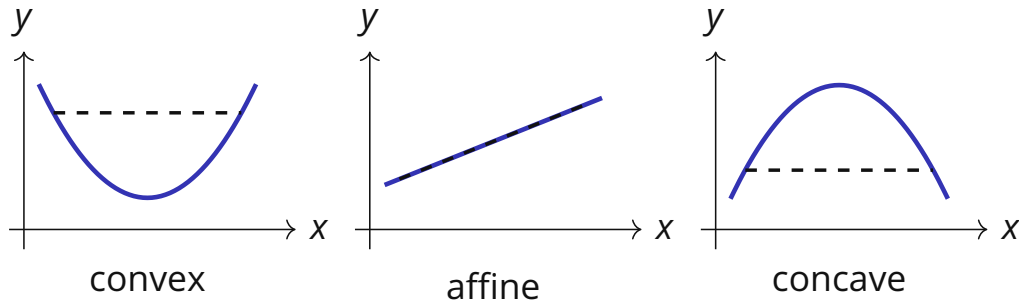


Figure 3 · Secants of three graphs.

The dashed secant lies above the convex graph, on the affine graph, and below the concave graph.

Degree-one homogeneity rules out strict concavity

Homogeneity of degree one is incompatible with strict concavity

Let D be a convex cone containing a nonzero x , and let $f : D \rightarrow \mathbb{R}$ be homogeneous of **degree one**. Choose $y = 2x$. Then $x \neq y$, while

$$\begin{aligned} f\left(\frac{1}{2}x + \frac{1}{2}y\right) &= f\left(\frac{3}{2}x\right) = \frac{3}{2}f(x), \\ \frac{1}{2}f(x) + \frac{1}{2}f(y) &= \frac{1}{2}f(x) + \frac{1}{2}f(2x) = \frac{3}{2}f(x). \end{aligned}$$

The two sides of the strict-concavity inequality are therefore equal for two distinct points, so strict concavity fails. A degree-one homogeneous function on a nontrivial convex cone is therefore never strictly concave. The same calculation rules out strict convexity.

Source: [Sydsæter et al., *Further Mathematics for Economic Analysis*, 2nd ed., §2.3, Problem 6\(c\)](#); the $x, 2x$ argument follows the Student's Manual solution to that problem.

Convexity from the definition

Let $f(x) = x^2$. To verify convexity, compute

$$\begin{aligned} & \lambda x^2 + (1 - \lambda)y^2 - (\lambda x + (1 - \lambda)y)^2 \\ &= \lambda(1 - \lambda)x^2 - 2\lambda(1 - \lambda)xy + \lambda(1 - \lambda)y^2 \\ &= \lambda(1 - \lambda)(x - y)^2 \geq 0. \end{aligned}$$

Therefore

$$(\lambda x + (1 - \lambda)y)^2 \leq \lambda x^2 + (1 - \lambda)y^2.$$

If $x \neq y$ and $\lambda \in (0, 1)$, the inequality is strict. Hence x^2 is strictly convex.

Concavity from the definition

Let $f(x) = \sqrt{x}$ on $[0, \infty)$. For $x, y \geq 0$ and $\lambda \in [0, 1]$,

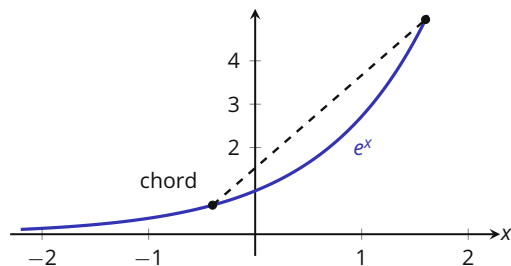
$$\begin{aligned}\lambda x + (1 - \lambda)y - (\lambda\sqrt{x} + (1 - \lambda)\sqrt{y})^2 &= \lambda(1 - \lambda)x - 2\lambda(1 - \lambda)\sqrt{xy} + \lambda(1 - \lambda)y \\ &= \lambda(1 - \lambda)(\sqrt{x} - \sqrt{y})^2 \geq 0.\end{aligned}$$

Both $\sqrt{\lambda x + (1 - \lambda)y}$ and $\lambda\sqrt{x} + (1 - \lambda)\sqrt{y}$ are nonnegative and $t \mapsto \sqrt{t}$ is increasing, so

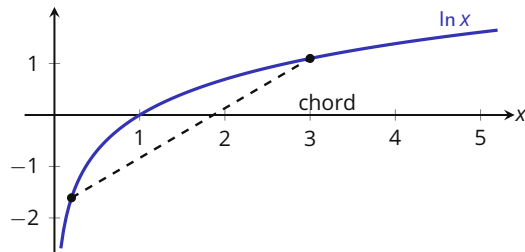
$$\sqrt{\lambda x + (1 - \lambda)y} \geq \lambda\sqrt{x} + (1 - \lambda)\sqrt{y}.$$

If $x \neq y$ and $\lambda \in (0, 1)$, the difference above is positive and the inequality is strict. Hence $\sqrt{x}$ is strictly concave: x^2 bends upward, its inverse $\sqrt{x}$ bends downward.

Exponential and logarithm



e^x is strictly convex: every chord lies above the graph.



$\ln x$ is strictly concave on $(0, \infty)$: every chord lies below the graph.

The two graphs are reflections of each other in the line $y = x$. The second-order test of the fourth section turns $(e^x)'' = e^x > 0$ and $(\ln x)'' = -1/x^2 < 0$ into these two statements; the exercise on weighted arithmetic and geometric means below uses the concavity of $\ln$.

Affine functions: boundary case

Let $f(x) = a^\top x + b$. Then

$$\begin{aligned}f(\lambda x + (1 - \lambda)y) &= a^\top (\lambda x + (1 - \lambda)y) + b \\&= \lambda a^\top x + (1 - \lambda)a^\top y + b \\&= \lambda f(x) + (1 - \lambda)f(y).\end{aligned}$$

The convexity inequality and the concavity inequality both hold with equality.

Conversely, if $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is both convex and concave, the two inequalities force equality for every x, y and $\lambda \in [0, 1]$. For $g = f - f(0)$, this equality gives $g(x + y) = g(x) + g(y)$ and $g(tx) = tg(x)$ for every $t \in \mathbb{R}$; hence $g(x) = a^\top x$ and f is affine.

Thus **a function is affine if and only if it is both convex and concave.**

In particular, when $b = 0$, $f(x) = a^\top x$ is linear; hence every linear function is both convex and concave.

Sums and affine substitutions

Let $S \subseteq \mathbb{R}^n$ be convex.

Nonnegative sums. If $f_1, \dots, f_m : S \rightarrow \mathbb{R}$ are convex and $\alpha_1, \dots, \alpha_m \geq 0$, then $\sum_i \alpha_i f_i$ is convex. With concave f_i , the sum is concave.

Affine substitution. If f is convex on a convex $D \subseteq \mathbb{R}^m$, then $g(x) = f(Ax + b)$ is convex on $\{x \in \mathbb{R}^n : Ax + b \in D\}$. With concave f , g is concave.

For the sum, apply each convexity inequality and use $\alpha_i \geq 0$:

$$\sum_i \alpha_i f_i(\lambda x + (1 - \lambda)y) \leq \sum_i \alpha_i [\lambda f_i(x) + (1 - \lambda)f_i(y)] = \lambda \sum_i \alpha_i f_i(x) + (1 - \lambda) \sum_i \alpha_i f_i(y).$$

For the substitution, $A(\lambda x + (1 - \lambda)y) + b = \lambda(Ax + b) + (1 - \lambda)(Ay + b)$, so $g(\lambda x + (1 - \lambda)y) \leq \lambda g(x) + (1 - \lambda)g(y)$. A negative weight reverses the inequality: f is convex iff $-f$ is concave.

Further reading: [Boyd and Vandenberghe, *Convex Optimization*, §§3.2.1, 3.2.2.](#)

Convexity without a Hessian

Let

$$f(x_1, x_2) = (x_1 + x_2 - 1)^2 + (2x_1 - x_2)^2.$$

Show that f is convex on $\mathbb{R}^2$ using only the convexity of $t \mapsto t^2$, the affine substitution rule, and the sum rule. Then decide whether $g(x_1, x_2) = (x_1 + x_2 - 1)^2 - (2x_1 - x_2)^2$ is convex.

Convexity without a Hessian

The maps $x \mapsto x_1 + x_2 - 1$ and $x \mapsto 2x_1 - x_2$ are affine and $t \mapsto t^2$ is convex, so both squares are convex functions of x by the affine substitution rule. Their sum, with weights $1, 1 \geq 0$, is convex by the sum rule. The Hessian test of the second-order section gives the same answer, with $H_f = 2 \begin{bmatrix} 5 & -1 \\ -1 & 2 \end{bmatrix}$.

For g , the second square enters with weight -1 , so the sum rule gives no conclusion. Along $x_1 = x_2 = t$, $g = (2t - 1)^2 - t^2 = 3t^2 - 4t + 1$ is convex in t ; along $x_1 = -x_2 = t$, $g = 1 - 9t^2$ is concave in t . By the affine substitution rule a convex g would be convex along every line, so g is neither convex nor concave.

Jensen's inequality

Let $S \subseteq \mathbb{R}^n$ be convex and $f : S \rightarrow \mathbb{R}$ convex. For $x_1, \dots, x_m \in S$ and $\lambda_i \geq 0$ with $\sum_i \lambda_i = 1$,

$$f\left(\sum_{i=1}^m \lambda_i x_i\right) \leq \sum_{i=1}^m \lambda_i f(x_i).$$

For a concave function, the inequality is reversed.

If f is strictly convex, every $\lambda_i > 0$, and the points x_i are not all equal, the inequality is strict. Induction on m : write $\sum_i \lambda_i x_i = \lambda_1 x_1 + (1 - \lambda_1)z$ with z the weighted average of $x_2, \dots, x_m$; if $x_1 \neq z$ the two-point inequality is strict, and if $x_1 = z$ the points $x_2, \dots, x_m$ cannot all be equal, so the claim for $m - 1$ points applies to them. The concave case is symmetric.

Video: [MIT OpenCourseWare, Jensen's inequality](#).

Jensen for random vectors

From finite mixtures to expectations

If X takes values in S , the expectations below exist, and $\mathbb{E}[X] \in S$, then

$$f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)]$$

for convex f , with the reverse inequality for concave f .

Further reading: [Boyd and Vandenberghe, Jensen's inequality](#).

Video: [Stanford EE364A, Lecture 3: convex functions, Hessian tests, epigraphs and Jensen's inequality](#).

Jensen inequality

Jensen's inequality for a two-point distribution

Let $f(x) = x^2$ and let X take values 1 and 3 with probability $1/2$ each.

FUNCTION AT THE MEAN

$$\mathbb{E}[X] = \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 3 = 2,$$

$$f(\mathbb{E}[X]) = f(2) = 4.$$

MEAN OF THE FUNCTION

$$\begin{aligned}\mathbb{E}[f(X)] &= \frac{1}{2}f(1) + \frac{1}{2}f(3) \\ &= \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 9 \\ &= 5.\end{aligned}$$

$$\text{Hence } f(\mathbb{E}[X]) = 4 < 5 = \mathbb{E}[f(X)].$$

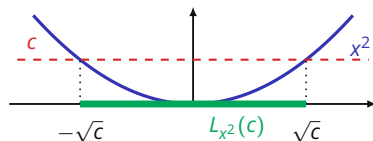
Lower and upper contour sets

Let $S \subseteq \mathbb{R}^n$ be convex, $f : S \rightarrow \mathbb{R}$, and $c \in \mathbb{R}$. The lower and upper contour sets of f at level c are

$$L_f(c) = \{x \in S : f(x) \leq c\}, \quad U_f(c) = \{x \in S : f(x) \geq c\};$$

the level set of the Functions section is $L_f(c) \cap U_f(c)$.

Contour sets. f convex $\Rightarrow$ every $L_f(c)$ is convex;
 f concave $\Rightarrow$ every $U_f(c)$ is convex.



$L_{x^2}(c) = [-\sqrt{c}, \sqrt{c}]$ is an interval.

For $x, y \in L_f(c)$: $f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y) \leq c$.

The converse fails: x^3 has the interval $(-\infty, c^{1/3}]$ as every lower contour set, yet it is not convex on $\mathbb{R}$.

The quasiconvexity section starts from this weaker property.

Epigraph & hypograph

Epigraph and hypograph tests

Epigraph.

$$\text{epi } f = \{(x, t) \in S \times \mathbb{R} : t \geq f(x)\}.$$



$\text{epi } f$ for $f(x) = x^2$.

For convex $S \subseteq \mathbb{R}^n$ and $f : S \rightarrow \mathbb{R}$:

$$f \text{ convex} \Leftrightarrow \text{epi } f \text{ convex}, \quad f \text{ concave} \Leftrightarrow \text{hypo } f \text{ convex}.$$

Function convexity is convexity of the epigraph, a set one dimension higher; since f is convex iff $-f$ is concave, the two statements are mirror images.

Hypograph.

$$\text{hypo } f = \{(x, t) \in S \times \mathbb{R} : t \leq f(x)\}.$$



$\text{hypo } g$ for $g(x) = 2 - 0.8x^2$.

Epigraph of x^2

Apply the epigraph test

Describe $\text{epi } f$ for $f(x) = x^2$, and verify directly that it is a convex subset of $\mathbb{R}^2$, as the epigraph test predicts for a convex f .

Epigraph of x^2

The region above the parabola

$$\text{epi} f = \{(x, s) \in \mathbb{R}^2 : s \geq x^2\}.$$

Take $(x_1, s_1), (x_2, s_2)$ in this set and $t \in [0, 1]$. Since $s_i \geq x_i^2$ and x^2 is convex,

$$ts_1 + (1 - t)s_2 \geq tx_1^2 + (1 - t)x_2^2 \geq (tx_1 + (1 - t)x_2)^2,$$

so the convex combination of the two points again lies above the parabola. The epigraph is convex, in agreement with the epigraph test.

Convex functions and Jensen's inequality

- (a) Use the definition to show that $f(x) = |x|$ is convex on $\mathbb{R}$.
- (b) Use Jensen's inequality to prove

$$\left(\frac{x+y}{2}\right)^2 \leq \frac{x^2+y^2}{2}.$$

When does equality hold?

Convex functions and Jensen's inequality

The triangle inequality gives

$$|tx + (1 - t)y| \leq t|x| + (1 - t)|y|, \quad 0 \leq t \leq 1,$$

which is the convexity inequality for $|x|$. For $q(z) = z^2$, Jensen with weights $\frac{1}{2}, \frac{1}{2}$ gives the requested inequality; equivalently,

$$\frac{x^2 + y^2}{2} - \left(\frac{x + y}{2}\right)^2 = \frac{(x - y)^2}{4} \geq 0,$$

with equality exactly when $x = y$.

Pointwise suprema and infima

The largest of a family of convex functions is convex

Let $S \subseteq \mathbb{R}^n$ be convex and let $f_\theta : S \rightarrow \mathbb{R}$ be convex for every θ in an index set Θ . If

$$g(x) = \sup_{\theta \in \Theta} f_\theta(x)$$

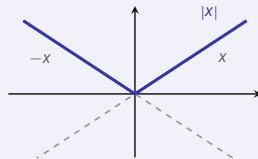
is finite on S , then g is convex. For each θ ,

$$f_\theta(\lambda x + (1 - \lambda)y) \leq \lambda f_\theta(x) + (1 - \lambda)f_\theta(y) \leq \lambda g(x) + (1 - \lambda)g(y),$$

and taking the supremum over θ on the left gives the convexity inequality for g . With concave f_θ , the pointwise infimum is concave where finite.

$|x| = \max\{x, -x\}$ is the maximum of two affine functions, hence convex, though it has no derivative at 0. The same rule is behind the profit function of the convex-sets section (next page).

Further reading: [Boyd and Vandenberghe, §3.2.3](#).



Maxima of affine functions

A piecewise linear convex function and the profit function

$f(x_1, x_2) = \max\{x_1, x_2, 0\}$ is the pointwise maximum of the three affine functions $x \mapsto x_1$, $x \mapsto x_2$, $x \mapsto 0$, hence convex on $\mathbb{R}^2$. Its graph consists of three planar pieces meeting along the rays $x_1 = x_2 \geq 0$, $x_1 = 0 \geq x_2$, and $x_2 = 0 \geq x_1$, where f has no gradient. A Hessian test says nothing here; the supremum rule settles convexity at once.

For a production set $Y \subseteq \mathbb{R}^n$, the profit function

$$\pi(p) = \sup_{y \in Y} p \cdot y$$

is the supremum over $y \in Y$ of the linear functions $p \mapsto p \cdot y$. By the rule, π is convex on the set of prices where it is finite, whatever the shape of Y ; this is the claim made on the production sets page.

Minimizing out a variable

Partial minimization of a jointly convex function

Let $C \subseteq \mathbb{R}^n \times \mathbb{R}^m$ be convex and let $f : C \rightarrow \mathbb{R}$ be convex in (x, y) jointly. Write $C_x = \{y : (x, y) \in C\}$ and suppose

$$g(x) = \inf_{y \in C_x} f(x, y)$$

is finite for every x in $X = \{x : C_x \neq \emptyset\}$, the image of C under $(x, y) \mapsto x$. Then g is convex on the convex set X . Take $x_1, x_2 \in X$, $\lambda \in [0, 1]$, and $\varepsilon > 0$. Choose $y_i \in C_{x_i}$ with $f(x_i, y_i) \leq g(x_i) + \varepsilon$. Convexity of C puts $\lambda(x_1, y_1) + (1 - \lambda)(x_2, y_2)$ in C , so $\lambda y_1 + (1 - \lambda)y_2 \in C_{\lambda x_1 + (1 - \lambda)x_2}$ and

$$g(\lambda x_1 + (1 - \lambda)x_2) \leq f(\lambda(x_1, y_1) + (1 - \lambda)(x_2, y_2)) \leq \lambda f(x_1, y_1) + (1 - \lambda)f(x_2, y_2) \leq \lambda g(x_1) + (1 - \lambda)g(x_2) + \varepsilon.$$

Letting $\varepsilon \downarrow 0$ gives the convexity inequality; for a jointly concave f , the supremum over y is concave. Without joint convexity a pointwise infimum of convex functions need not be convex: see the double well on the convexification page. Further reading: [Boyd and Vandenberghe, §3.2.5](#).

Composition rules

Curvature and monotonicity of the outer function

Let $g : S \rightarrow \mathbb{R}$ on a convex S , and let $h : I \rightarrow \mathbb{R}$ on an interval $I \supseteq g(S)$.

Outer h	Inner g	Composite $h \circ g$
convex, nondecreasing	convex	convex
convex, nonincreasing	concave	convex
concave, nondecreasing	concave	concave
concave, nonincreasing	convex	concave

First row: $g(\lambda x + (1 - \lambda)y) \leq \lambda g(x) + (1 - \lambda)g(y)$ and h nondecreasing give $h(g(\lambda x + (1 - \lambda)y)) \leq h(\lambda g(x) + (1 - \lambda)g(y)) \leq \lambda h(g(x)) + (1 - \lambda)h(g(y))$, the second step by convexity of h ; the point $\lambda g(x) + (1 - \lambda)g(y)$ lies in I because I is an interval. The other rows reverse one or both inequalities. Monotonicity alone is too weak: $g(x) = x$ is concave and $h = \exp$ is increasing, yet e^x is strictly convex, so it is not concave; row three also needs h concave. The quasiconcavity section shows what an increasing h does preserve.

Further reading: [Boyd and Vandenberghe, §3.2.4](#).

The perspective of a convex function

Scaling a convex function into a degree-one homogeneous one

For $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the perspective is

$$g(x, t) = tf(x/t), \quad x \in \mathbb{R}^n, t > 0.$$

It is homogeneous of degree one in (x, t) and convex whenever f is: for $(x_1, t_1), (x_2, t_2), \lambda \in [0, 1]$, and $T = \lambda t_1 + (1 - \lambda)t_2 > 0$,

$$\frac{\lambda x_1 + (1 - \lambda)x_2}{T} = \frac{\lambda t_1}{T} \frac{x_1}{t_1} + \frac{(1 - \lambda)t_2}{T} \frac{x_2}{t_2}$$

is a convex combination of x_1/t_1 and x_2/t_2 , so convexity of f and multiplication by T give

$$g(\lambda x_1 + (1 - \lambda)x_2, T) \leq \lambda t_1 f(x_1/t_1) + (1 - \lambda)t_2 f(x_2/t_2) = \lambda g(x_1, t_1) + (1 - \lambda)g(x_2, t_2).$$

With concave f , g is concave. Example: $f(x) = x^2$ gives $g(x, t) = x^2/t$, convex on $\mathbb{R} \times (0, \infty)$.

Further reading: [Boyd and Vandenberghe, §3.2.6](#).

Randomization and cost

Randomization produces convex costs

Let $C(\mu)$ be the minimum cost of producing distribution μ , assume the minima are attained, and suppose randomization across procedures is feasible.

Randomizing between procedures that minimize cost for μ and ν with probabilities α and $1 - \alpha$ produces $\alpha\mu + (1 - \alpha)\nu$ at expected cost $\alpha C(\mu) + (1 - \alpha)C(\nu)$. Hence:

$$C(\alpha\mu + (1 - \alpha)\nu) \leq \alpha C(\mu) + (1 - \alpha)C(\nu).$$

Randomization produces the convexity inequality for C by mixing procedures.

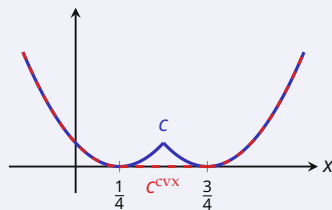
Finite randomization convexifies feasible outcomes and costs

General convexification

If deterministic procedures generate outcomes in A , finite randomization generates every point in $\text{conv}(A)$. For a deterministic cost $c : A \rightarrow \mathbb{R}$ that is bounded below, the infimum expected cost of generating $x \in \text{conv}(A)$ is

$$c^{\text{cvx}}(x) = \inf \left\{ \sum_{i=1}^m \lambda_i c(x_i) : m \geq 1, x_i \in A, \lambda_i \geq 0, \right. \\ \left. \sum_{i=1}^m \lambda_i = 1, \sum_{i=1}^m \lambda_i x_i = x \right\}.$$

This convex envelope is the largest convex function on $\text{conv}(A)$ that does not exceed c on A .



Dashed: the convex envelope of the double well c , the smaller of $(x - \frac{1}{4})^2$ and $(x - \frac{3}{4})^2$.

The concave envelope is the smallest concave majorant

Concave envelopes

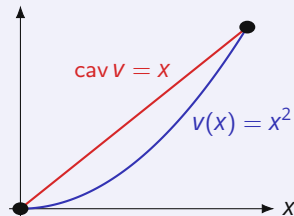
Let $D \subseteq \mathbb{R}^n$ be convex and let $v : D \rightarrow \mathbb{R}$ be bounded above. Its smallest concave majorant is

$$\text{cav } v(x) = \sup_{\substack{m \geq 1, x_i \in D, \lambda_i \geq 0 \\ \sum_i \lambda_i = 1, \sum_i \lambda_i x_i = x}} \sum_i \lambda_i v(x_i).$$

On $D = [0, 1]$, let $v(x) = x^2$. The affine function $h(x) = x$ majorizes v . If g is any concave majorant of v , then

$$g(x) \geq (1-x)g(0) + xg(1) \geq x = h(x).$$

Hence $\text{cav } v = h$, so $\text{cav } v(x) = x$.



Restriction to a line

Convexity in $\mathbb{R}^n$ is convexity along every line

Let $S \subseteq \mathbb{R}^n$ be convex and $f : S \rightarrow \mathbb{R}$. Then f is convex iff for every $x \in S$ and $v \in \mathbb{R}^n$ the function of one variable

$$\varphi(t) = f(x + tv), \quad t \in \{t \in \mathbb{R} : x + tv \in S\},$$

is convex on its domain, which is an interval.

If f is convex, φ is convex by the affine substitution rule. Conversely, given $x, y \in S$ and $\lambda \in [0, 1]$, take $v = y - x$: the domain of φ contains $[0, 1]$, and

$$f(x + \lambda(y - x)) = \varphi((1 - \lambda) \cdot 0 + \lambda \cdot 1) \leq (1 - \lambda)\varphi(0) + \lambda\varphi(1) = (1 - \lambda)f(x) + \lambda f(y).$$

For twice differentiable f , the chain rule gives $\varphi''(t) = v^\top H_f(x + tv) v$; the next two sections turn this into the gradient and Hessian tests, one direction v at a time.

Further reading: [Boyd and Vandenberghe, §3.1.1](#).

Section 3

First-order characterizations

First-order characterization in $\mathbb{R}^n$

Let $S \subseteq \mathbb{R}^n$ be **open and convex**, and let $f : S \rightarrow \mathbb{R}$ be differentiable. Then

Convex iff.

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x) \quad \forall x, y \in S.$$

Concave iff.

$$f(y) \leq f(x) + \nabla f(x)^\top (y - x) \quad \forall x, y \in S.$$

For $n = 1$, these are the familiar tangent-line inequalities.

Video: [Mike, the Mathematician, *First Order Conditions for Convexity*](#).

Gradient monotonicity

From two tangent inequalities to one inner product

Let $S \subseteq \mathbb{R}^n$ be open and convex, and let $f : S \rightarrow \mathbb{R}$ be differentiable and convex. Prove that

$$(\nabla f(x) - \nabla f(y))^{\top} (x - y) \geq 0 \quad \text{for all } x, y \in S.$$

Specialize the result to $f(z) = \frac{1}{2}z^{\top}Az$ on $\mathbb{R}^n$, where $A = A^{\top}$.

Gradient monotonicity

Add the two convexity inequalities

Convexity gives

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x), \quad f(x) \geq f(y) + \nabla f(y)^\top (x - y).$$

Adding and rearranging yields

$$(\nabla f(x) - \nabla f(y))^\top (x - y) \geq 0.$$

For $f(z) = \frac{1}{2}z^\top Az$, $\nabla f(z) = Az$, so the inequality is $(x - y)^\top A(x - y) \geq 0$. Since $x - y$ can be any $h \in \mathbb{R}^n$, this is $h^\top Ah \geq 0$ for every h , which is the condition $A \succeq 0$. In one variable, $(f'(x) - f'(y))(x - y) \geq 0$: the derivative of a differentiable convex function is nondecreasing.

First-order characterization: supporting tangent planes

For a convex function

$$f(u, v) = \frac{1}{2}(u^2 + v^2), \quad \bar{x} = (1, 0),$$

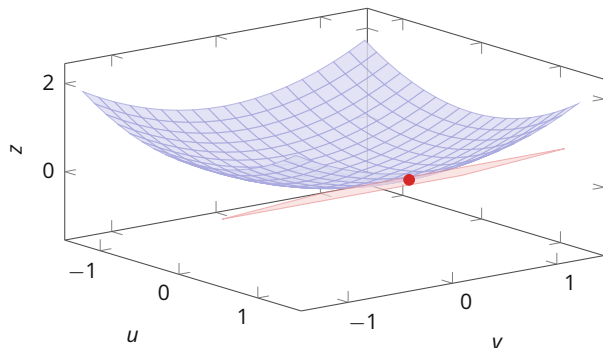
we have $f(\bar{x}) = \frac{1}{2}$ and $\nabla f(\bar{x}) = (1, 0)^\top$. The tangent plane is

$$L(u, v) = f(\bar{x}) + \nabla f(\bar{x})^\top \begin{bmatrix} u - 1 \\ v \end{bmatrix} = u - \frac{1}{2}.$$

Moreover,

$$f(u, v) - L(u, v) = \frac{1}{2}((u - 1)^2 + v^2) \geq 0.$$

Thus, for convex f , every tangent plane supports the graph from below. For concave f , every tangent plane supports the graph from above.



Blue: the graph $z = f(u, v)$. Red: the tangent plane $z = u - \frac{1}{2}$ at $\bar{x}$ (dot).

The graph of a convex function lies above its tangent plane.

Tangent test

CONVEX FUNCTION For $f(x) = x^2$ at $x_0 = 1$, the tangent line is

$$f(1) + f'(1)(x - 1) = 1 + 2(x - 1) = 2x - 1.$$

The graph stays above it:

$$x^2 - (2x - 1) = (x - 1)^2 \geq 0.$$

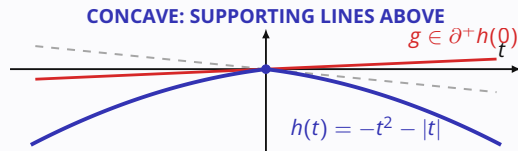
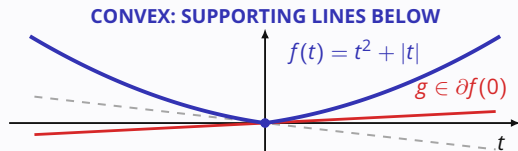
CONCAVE FUNCTION For $g(x) = \ln x$ at $x_0 = 1$, the tangent line is

$$g(1) + g'(1)(x - 1) = x - 1.$$

Concavity gives the global upper bound

$$\ln x \leq x - 1, \quad x > 0.$$

Subgradients & supergradients



First-order bounds without differentiability

Let $S \subseteq \mathbb{R}^n$ be open and convex, $x \in S$, and $f : S \rightarrow \mathbb{R}$.

Subdifferential for convex f .

$$\partial f(x) = \{g \in \mathbb{R}^n : f(y) \geq f(x) + g^\top (y - x), \forall y \in S\}.$$

Superdifferential for concave f .

$$\partial^+ f(x) = \{g \in \mathbb{R}^n : f(y) \leq f(x) + g^\top (y - x), \forall y \in S\}.$$

If f is differentiable at x . $\partial f(x) = \{\nabla f(x)\}$ (convex), $\partial^+ f(x) = \{\nabla f(x)\}$ (concave).

Video: Kiet, *Understanding Subgradients Using Examples*.

Subgradients of the absolute value function

Subdifferential of $|x|$

Let $f(x) = |x|$.

- (a) Find $\partial f(0)$.
- (b) Find $\partial f(x)$ for $x > 0$ and for $x < 0$.
- (c) For each $g \in \partial f(0)$, write the corresponding global affine lower bound.

Subgradients of the absolute value function

A whole interval of slopes at the kink

At 0, a number g is a subgradient exactly when

$$|y| \geq gy \quad \text{for every } y \in \mathbb{R}.$$

Positive y requires $g \leq 1$, while negative y requires $g \geq -1$. Hence

$$\partial f(0) = [-1, 1], \quad \partial f(x) = \begin{cases} \{1\}, & x > 0, \\ \{-1\}, & x < 0. \end{cases}$$

Each $g \in [-1, 1]$ gives the global lower bound $|y| \geq gy$.

Section 4

Second-order characterizations

Second-order test in 1D

Let $S \subseteq \mathbb{R}$ be an open interval and let $f : S \rightarrow \mathbb{R}$ be twice differentiable:

Weak inequalities.

$$f'' \geq 0 \text{ on } S \Leftrightarrow f \text{ convex on } S,$$

$$f'' \leq 0 \text{ on } S \Leftrightarrow f \text{ concave on } S.$$

Strict inequalities.

$$f'' > 0 \text{ on } S \Rightarrow f \text{ strictly convex on } S,$$

$$f'' < 0 \text{ on } S \Rightarrow f \text{ strictly concave on } S.$$

Note the directions of implications. The **weak** inequalities **characterize** convexity and concavity. The **strict** inequalities are **sufficient** but **not necessary** conditions.

Video: Joydeep Dutta, *Twice differentiable convex functions*.

Strict convexity with zero curvature at one point

Verify the definition. Fix $x \neq y$ and $\lambda \in (0, 1)$, and set $m = \lambda x + (1 - \lambda)y$. Then

$$\lambda x^2 + (1 - \lambda)y^2 - m^2 = \lambda(1 - \lambda)(x - y)^2 > 0.$$

Hence

$$m^4 < (\lambda x^2 + (1 - \lambda)y^2)^2 \leq \lambda x^4 + (1 - \lambda)y^4,$$

where the second inequality follows from

$$\lambda(1 - \lambda)(x^2 - y^2)^2 \geq 0.$$

Thus $f(m) < \lambda f(x) + (1 - \lambda)f(y)$: $f(x) = x^4$ is strictly convex by definition.

THE CURVATURE TEST IS ONE-WAY

$$f(x) = x^4, \quad f'(x) = 4x^3, \quad f''(x) = 12x^2.$$

At the origin,

$$f''(0) = 0.$$

$f'' > 0$ everywhere $\Rightarrow$ strict convexity,
but the reverse implication fails.

A strictly concave quadratic

Curvature and the global maximizer

Let

$$f(x) = 10x - x^2, \quad x \geq 0.$$

Solving $f'(x) = 0$ gives

$$10 - 2x = 0 \quad \Rightarrow \quad x^* = 5.$$

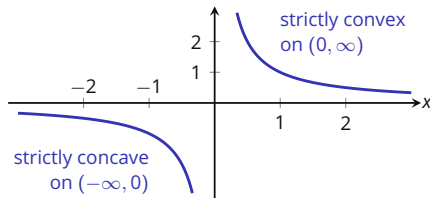
Because $f''(x) = -2 < 0$, the function is strictly concave on $\mathbb{R}_+$. Thus $x^* = 5$ is the unique global maximizer, with

$$f(5) = 10 \cdot 5 - 5^2 = 25.$$

The domain is part of the statement

CURVATURE CAN CHANGE ACROSS REGIONS

For $f(x) = x^3$ we get $f''(x) = 6x$, so f is strictly convex on $(0, \infty)$, strictly concave on $(-\infty, 0)$, and neither on $\mathbb{R}$, where f'' changes sign.



THE NATURAL DOMAIN CAN FAIL TO BE CONVEX

For $f(x) = 1/x$ we get $f''(x) = 2/x^3$, positive on $(0, \infty)$ and negative on $(-\infty, 0)$. Its natural domain skips the midpoint: $-1, 1 \in \mathbb{R} \setminus \{0\}$ while $\frac{-1+1}{2} = 0 \notin \mathbb{R} \setminus \{0\}$.

Domain rule. Convexity and concavity are stated on a convex domain. For $f(x) = 1/x$, use the two separate intervals: strict concavity on $(-\infty, 0)$ and strict convexity on $(0, \infty)$.

Weighted arithmetic and geometric means

Let $x, y > 0$ and $0 < \lambda < 1$. Use differentiation and Jensen's inequality to prove

$$x^\lambda y^{1-\lambda} \leq \lambda x + (1 - \lambda)y.$$

Determine when equality holds.

Weighted arithmetic and geometric means

For $g(t) = \ln t$,

$$g''(t) = -\frac{1}{t^2} < 0,$$

so g is strictly concave. Jensen's inequality gives

$$\ln(\lambda x + (1 - \lambda)y) \geq \lambda \ln x + (1 - \lambda) \ln y = \ln(x^\lambda y^{1-\lambda}).$$

Applying the increasing function $\exp$ proves the inequality. Since $0 < \lambda < 1$ and $\ln$ is strictly concave, equality holds exactly when $x = y$.

From Taylor curvature to a matrix sign test

In one variable the criterion is the sign of the number $f''(x)$. In $\mathbb{R}^n$ the second derivative is the Hessian matrix $H_f(x)$, and the second-order expansion from Differentiation says what its sign has to mean:

$$f(x+h) - f(x) = \nabla f(x)^\top h + \frac{1}{2} h^\top H_f(x) h + o(\|h\|^2).$$

The second-order term is

$$Q_x(h) := h^\top H_f(x) h,$$

a quadratic form whose representing matrix is the Hessian.

So the n -dimensional analogue of $f''(x) \geq 0$ is a statement about the sign of $Q_x(h)$ across directions h , which is what the matrix tests below decide.

Video: [Ketan Rajawat, First and second order conditions for convexity.](#)

Hessian signs across a domain

This section uses the Hessian to classify curvature *across a domain*.

Hessian criterion on a domain. Let $S \subseteq \mathbb{R}^n$ be open and convex, and let $f : S \rightarrow \mathbb{R}$ be twice differentiable. Then

$$f \text{ is convex on } S \Leftrightarrow h^\top H_f(x)h \geq 0 \quad \text{for every } x \in S \text{ and } h \in \mathbb{R}^n,$$

and

$$f \text{ is concave on } S \Leftrightarrow h^\top H_f(x)h \leq 0 \quad \text{for every } x \in S \text{ and } h \in \mathbb{R}^n.$$

Definiteness

For a twice differentiable function, second-order change in a direction h is governed by $h^\top H_f(x)h$. A **quadratic form** on $\mathbb{R}^n$ can be written $Q(x) = x^\top Ax$; take $A = A^\top$ because only the symmetric part affects $x^\top Ax$.

Positive definite

$$x^\top Ax > 0 \text{ for all } x \neq 0.$$

Negative definite

$$x^\top Ax < 0 \text{ for all } x \neq 0.$$

Positive semidefinite

$$x^\top Ax \geq 0 \text{ for all } x.$$

Negative semidefinite

$$x^\top Ax \leq 0 \text{ for all } x.$$

Indefinite

$$x^\top Ax > 0 > y^\top Ay \text{ for some } x, y.$$

An indefinite form is positive in some directions and negative in others.

Covariance matrices and portfolio variance

Every covariance matrix is positive semidefinite. Let $\Sigma = \text{Cov}(X)$. For every $a \in \mathbb{R}^n$,

$$a^\top \Sigma a = \text{Var}(a^\top X) \geq 0.$$

Hence $\Sigma \succeq 0$. It is positive definite exactly when

$$\text{Var}(a^\top X) > 0 \quad \text{for every } a \neq 0.$$

A DEPENDENT PAIR If $X = (Z, 2Z)^\top$, then

$$\Sigma = \text{Var}(Z) \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}.$$

For $a = (2, -1)^\top \neq 0$,

$$a^\top X = 0, \quad a^\top \Sigma a = 0.$$

This nonzero linear combination has zero variance, so $\Sigma \succeq 0$ but $\Sigma \not\succ 0$.

Video: [Brunei Math Club](#), *Why every covariance matrix is positive semidefinite*.

Homogeneity and sign along rays

Scaling a quadratic form

For $Q(x) = x^\top Ax$ and any scalar $\lambda \in \mathbb{R}$,

$$Q(\lambda x) = (\lambda x)^\top A(\lambda x) = \lambda^2 x^\top Ax = \lambda^2 Q(x).$$

Thus Q is homogeneous of degree two; in particular, $Q(0) = 0$ and $Q(-x) = Q(x)$. If $x \neq 0$ and $\lambda \neq 0$, then $\lambda^2 > 0$, so

$$\text{sign } Q(\lambda x) = \text{sign } Q(x).$$

On each line through the origin, Q is positive away from the origin, negative away from the origin, or identically zero.

Further reading: [Boyd and Vandenberghe, quadratic-form slides](#).

Geometry: an indefinite quadratic form

For

$$Q(x_1, x_2) = x_1(x_1 - 4x_2),$$

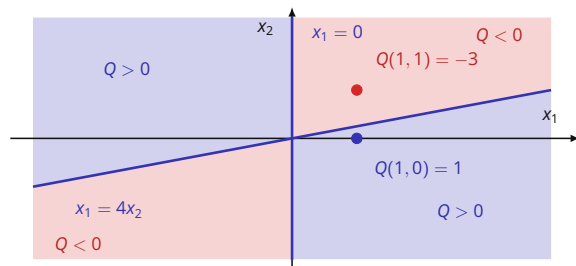
the zero directions are $x_1 = 0$ and $x_1 = 4x_2$.

Moreover,

$$Q(1, 0) = 1 > 0, \quad Q(1, 1) = -3 < 0.$$

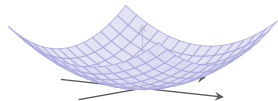
The zero lines separate positive and negative directional sectors, so Q is indefinite.

Definiteness concerns the sign across all directions, not the value at one vector.



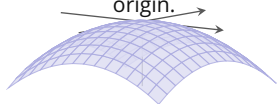
Each open sector has a constant sign.

Geometry of quadratic forms



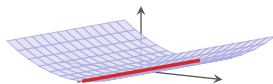
POSITIVE DEFINITE

$Q = x_1^2 + x_2^2$: a bowl, $Q > 0$ off the origin.



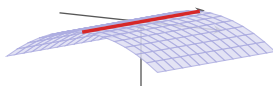
NEGATIVE DEFINITE

$Q = -x_1^2 - x_2^2$: the bowl turned over.



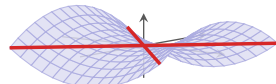
POSITIVE SEMIDEFINITE

$Q = x_1^2$: a trough, $Q = 0$ on the line $x_1 = 0$ (red).



NEGATIVE SEMIDEFINITE

$Q = -x_1^2$: a ridge, $Q = 0$ on $x_1 = 0$ (red).



INDEFINITE

$Q = x_1^2 - x_2^2$: a saddle, $Q = 0$ on $x_1 = \pm x_2$ (red).

Definite: $Q \neq 0$ in every nonzero direction.

Semidefinite but not definite: Q has one weak sign and vanishes along some nonzero directions (red); this is the "semi".

Indefinite: Q takes both signs.

Examples of definiteness

POSITIVE DEFINITE

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \quad x^\top A x = 2x_1^2 + x_2^2 > 0$$

for every $x \neq 0$.

POSITIVE SEMIDEFINITE

$$B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad x^\top B x = (x_1 + x_2)^2 \geq 0.$$

But for $x = (1, -1)^\top$, $x^\top B x = 0$.

NEGATIVE DEFINITE

$$C = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}, \quad x^\top C x = -x_1^2 - 2x_2^2 < 0$$

for every $x \neq 0$.

INDEFINITE

$$D = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad x^\top D x = x_1^2 - x_2^2.$$

It is positive at $(1, 0)^\top$ and negative at $(0, 1)^\top$.

Definiteness and eigenvalues

Let $A = A^T$. Its eigenvalues are real, and their signs give an equivalent definiteness test:

Eigenvalue sign test.

A positive definite	$\Leftrightarrow$	$\lambda_i(A) > 0$ for every i ,
A positive semidefinite	$\Leftrightarrow$	$\lambda_i(A) \geq 0$ for every i ,
A negative definite	$\Leftrightarrow$	$\lambda_i(A) < 0$ for every i ,
A negative semidefinite	$\Leftrightarrow$	$\lambda_i(A) \leq 0$ for every i .

A symmetric matrix is indefinite exactly when it has at least one positive and at least one negative eigenvalue. Eigenvalues and principal-minor tests are alternative ways to determine the same sign structure.

Further reading: [Boyd and Vandenberghe, quadratic-form slides](#).

Principal submatrices and principal minors

Let A be an $n \times n$ matrix and let $I \subseteq \{1, \dots, n\}$ contain k indices.

Principal submatrix of order k . $A[I, I]$ keeps the rows indexed by I and the same columns indexed by I .

Principal minor of order k .

$M_I := \det A[I, I]$, one for each of the $\binom{n}{k}$ index sets.

A **leading** principal submatrix uses $I = \{1, \dots, k\}$, with determinant $D_k = \det A[I, I]$.

WORKED INDEX SET: $I = \{1, 3\}$

$$A = \begin{bmatrix} 4 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 0 & 5 \end{bmatrix} \Rightarrow A[I, I] = \begin{bmatrix} 4 & 2 \\ 2 & 5 \end{bmatrix},$$

so $M_{\{1,3\}} = 20 - 4 = 16$: row 2 and column 2 are removed together.

For $n = 3$ the index sets are $\{1\}, \{2\}, \{3\}; \{1, 2\}, \{1, 3\}, \{2, 3\}; \{1, 2, 3\}$: seven principal minors, of which D_1, D_2, D_3 are leading. An $n \times n$ matrix has $2^n - 1$ principal minors and n leading ones.

Video: [The Complete Guide, Sylvester's criterion](#), on the leading minors D_k .

Definiteness via minors

Let $A = A^\top$, let D_k be its leading principal minors, and let $M_I = \det A[I, I]$.

Definite: n leading minors.

$$A \succ 0 \Leftrightarrow D_k > 0 \quad (k = 1, \dots, n),$$

$$A \prec 0 \Leftrightarrow (-1)^k D_k > 0 \quad (k = 1, \dots, n).$$

Semidefinite: all principal minors.

$$A \succeq 0 \Leftrightarrow M_I \geq 0 \quad (I \neq \emptyset),$$

$$A \preceq 0 \Leftrightarrow (-1)^{|I|} M_I \geq 0 \quad (I \neq \emptyset).$$

Strict sign + leading minors

tests definiteness;

weak sign + every principal minor tests semidefiniteness.

For negative semidefiniteness, odd-order principal minors are nonpositive, even-order principal minors are nonnegative, and zeros are allowed.

Leading minors do not test semidefiniteness

When zeros appear in the leading sequence, a wrongly signed principal minor can sit on an omitted index set.

LEADING MINORS LOOK ADMISSIBLE

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 0 & 0 \\ 2 & 0 & 2 \end{bmatrix},$$

$$D_1 = 1 > 0, \quad D_2 = 0, \quad D_3 = \det A = 0,$$

so every leading principal minor is nonnegative.

The omitted index set $I = \{1, 3\}$.

$$A[I, I] = \begin{bmatrix} 1 & 2 \\ 2 & 2 \end{bmatrix}, \quad M_{\{1,3\}} = 2 - 4 = -2 < 0.$$

Positive semidefiniteness requires *every* principal minor to be nonnegative, so this one minor rules it out.

THE SAME FAILURE IN $x^T A x$

With $Q(x) = x_1^2 + 4x_1x_3 + 2x_3^2$, $Q(1, 0, 1) = 7 > 0$ and $Q(1, 0, -1) = -1 < 0$, so A is indefinite: the omitted minor and the two directions detect one sign failure.

Video: [Arizona Math Camp, Lecture 29\(C\): quadratic forms: determinant conditions for definiteness.](#)

The 2×2 case

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{bmatrix},$$

$$\det A = a_{11}a_{22} - a_{12}^2.$$

$$\det A < 0 \quad A \text{ indefinite}$$

$$\det A > 0, a_{11} > 0 \quad A \succ 0$$

$$\det A > 0, a_{11} < 0 \quad A \prec 0$$

$$\det A \geq 0, a_{11} \geq 0, a_{22} \geq 0 \quad A \succeq 0$$

$$\det A \geq 0, a_{11} \leq 0, a_{22} \leq 0 \quad A \preceq 0$$

The definite rows need only a_{11} , because $\det A > 0$ forces $a_{11}a_{22} > a_{12}^2 \geq 0$, so the two diagonal entries carry the same strict sign. That implication disappears at $\det A = 0$, so the semidefinite rows check both diagonal entries, which are the principal minors of order one.

The 3×3 case

For a symmetric matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{12} & a_{22} & a_{23} \\ a_{13} & a_{23} & a_{33} \end{bmatrix},$$

the leading principal minors are

$$D_1 = a_{11},$$

$$D_2 = \det \begin{bmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{bmatrix}, \quad D_3 = \det A.$$

Hence Sylvester's criterion specializes to

$$A \succ 0 \Leftrightarrow D_1 > 0, D_2 > 0, D_3 > 0,$$

$$A \prec 0 \Leftrightarrow D_1 < 0, D_2 > 0, D_3 < 0.$$

Definiteness also fixes the sign of the other four principal minors: if $A \succ 0$ all seven are positive, and if $A \prec 0$ each minor of order k has sign $(-1)^k$. Testing A still takes only the three leading minors D_1, D_2, D_3 ; the remaining four follow.

Portfolio risk matrix

From $\Sigma \succeq 0$ to $\Sigma \succ 0$ for one covariance matrix

Variance already gives $\Sigma \succeq 0$ for every covariance matrix; the minors settle the strict case for a given one. For three asset returns with

$$\Sigma = \begin{bmatrix} 4 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{bmatrix}, \quad w^T \Sigma w = \text{Var}(w^T X) :$$

LEADING PRINCIPAL MINORS

$$D_1 = 4, \quad D_2 = \begin{vmatrix} 4 & 1 \\ 1 & 3 \end{vmatrix} = 11,$$

$$D_3 = \det \Sigma = 18.$$

All positive, so $\Sigma \succ 0$: every portfolio with $w \neq 0$ carries strictly positive variance.

THE OTHER FOUR PRINCIPAL MINORS

$$\text{Order 1: } 3, 2; \quad \text{order 2: } \begin{vmatrix} 3 & 1 \\ 1 & 2 \end{vmatrix} = 5, \quad \begin{vmatrix} 4 & 0 \\ 0 & 2 \end{vmatrix} = 8.$$

All are positive, as $\Sigma \succ 0$ requires, and none entered the test above.

Workflow for definiteness tests

Let $A = A^\top$. The minors route proceeds in three steps.

1. LEADING MINORS

Match

$+, +, \dots, +$

or

$-, +, -, \dots$

If one matches, A is definite.



2. ALL PRINCIPAL MINORS

If Step 1 fails, test

$$M_I \geq 0 \quad \forall I$$

and

$$(-1)^{|I|} M_I \geq 0 \quad \forall I.$$



3. CLASSIFY A weak sign pattern gives $A \succeq 0$ or $A \preceq 0$.

If neither holds, A is indefinite. Two vectors giving opposite signs certify this directly.

Eigenvalues provide a one-pass alternative: their signs classify every symmetric matrix directly. The Hessian criterion on the next page turns this matrix classification into a convexity or concavity classification.

Definiteness from principal minors

Classify each symmetric matrix using principal minors:

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}.$$

For each matrix, state whether the test above settles the answer. If it does not, inspect all principal minors or exhibit vectors with quadratic forms of opposite signs.

Definiteness from principal minors

MATRIX A

$$D_1 = 2 > 0, \quad D_2 = 3 > 0.$$

Thus $A \succ 0$.

MATRIX B Its leading determinant is zero. All principal minors are 1, 1, 0, so $B \succeq 0$, but $B \not\succ 0$.

MATRIX C $D_2 = 1 - 4 = -3 < 0$. Also $e_1^T C e_1 = 1 > 0$, while $(1, -1)C(1, -1)^T = -2 < 0$. Thus C is indefinite.

Second-order test in $\mathbb{R}^n$

Let $S \subseteq \mathbb{R}^n$ be open and convex, and let $f : S \rightarrow \mathbb{R}$ be twice continuously differentiable.

NECESSARY AND SUFFICIENT $\Leftrightarrow$

f convex on $S \Leftrightarrow H_f(x) \succeq 0$ for every $x \in S$,

f concave on $S \Leftrightarrow H_f(x) \preceq 0$ for every $x \in S$.

SUFFICIENT ONLY $\Rightarrow$

$H_f(x) \succ 0$ for every $x \in S \Rightarrow f$ strictly convex on S ,

$H_f(x) \prec 0$ for every $x \in S \Rightarrow f$ strictly concave on S .

Hessian test

STRICTLY CONVEX Let $f(x, y) = x^2 + xy + y^2$. Then

$$H_f = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

Leading minors are

$$D_1 = 2 > 0, \quad D_2 = 4 - 1 = 3 > 0.$$

Hence H_f is positive definite.

Video: [MIT OpenCourseWare, Positive Definite Matrices and Minima](#).

STRICTLY CONCAVE Let $g(x, y) = \ln x + \ln y$ on $\mathbb{R}_{++}^2$. Then

$$H_g = \begin{bmatrix} -1/x^2 & 0 \\ 0 & -1/y^2 \end{bmatrix}.$$

For $v \neq 0$,

$$v^\top H_g v = -\frac{v_1^2}{x^2} - \frac{v_2^2}{y^2} < 0.$$

Classifying functions with Hessians

Compute the gradient and Hessian, then classify each function on its full Euclidean domain:

$$f(x,y) = x^3 + y^2, \quad g(x,y,z) = 3x^2 + y^2 - 2xy + e^z.$$

Classifying functions with Hessians

GRADIENTS, HESSIANS, AND CLASSIFICATIONS

For the first function,

$$\nabla f(x, y) = \begin{bmatrix} 3x^2 \\ 2y \end{bmatrix},$$

$$H_f(x, y) = \begin{bmatrix} 6x & 0 \\ 0 & 2 \end{bmatrix}.$$

Its sign changes with x , so f is neither convex nor concave on $\mathbb{R}^2$.

For the second,

$$\nabla g(x, y, z) = \begin{bmatrix} 6x - 2y \\ 2y - 2x \\ e^z \end{bmatrix},$$

$$H_g(x, y, z) = \begin{bmatrix} 6 & -2 & 0 \\ -2 & 2 & 0 \\ 0 & 0 & e^z \end{bmatrix}.$$

Its leading principal minors are 6, 8, and $8e^z$, all positive. Thus H_g is positive definite and g is strictly convex on $\mathbb{R}^3$.

A quadratic lower contour set

Let

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad q(x) = x^\top A x,$$

and define

$$C = \{x \in \mathbb{R}^2 : q(x) \leq 1\}, \quad D = \{x \in \mathbb{R}^2 : q(x) \geq 1\}.$$

Compute ∇q and H_q ; show that q is strictly convex; prove that C is convex; and determine whether D is convex.

A quadratic lower contour set

Since $A = A^\top$,

$$\nabla q(x) = 2Ax, \quad H_q = 2A = \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}.$$

The leading principal minors are 4 and 12, so $H_q \succ 0$ and q is strictly convex. If $x, y \in C$ and $0 \leq \lambda \leq 1$,

$$q(\lambda x + (1 - \lambda)y) \leq \lambda q(x) + (1 - \lambda)q(y) \leq 1,$$

so C is convex. The set D is not convex: $(1, 0), (-1, 0) \in D$, but their midpoint is 0 and $q(0) = 0 < 1$.

A Hessian with rank one

Let

$$a = \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \quad f(x) = e^{a^\top x} = e^{x_1 - 2x_2}.$$

Use the chain rule to compute ∇f and H_f . Prove from the quadratic form $v^\top H_f(x)v$ that f is convex on $\mathbb{R}^2$. Is f strictly convex on $\mathbb{R}^2$?

A Hessian with rank one

The chain rule gives

$$\nabla f(x) = e^{a^\top x} a, \quad H_f(x) = e^{a^\top x} a a^\top = e^{x_1 - 2x_2} \begin{bmatrix} 1 & -2 \\ -2 & 4 \end{bmatrix}.$$

Hence, for every $v \in \mathbb{R}^2$,

$$v^\top H_f(x) v = e^{a^\top x} (a^\top v)^2 \geq 0,$$

so f is convex. It is not strictly convex on $\mathbb{R}^2$: for $x = (0, 0)$ and $y = (2, 1)$, $a^\top x = a^\top y = 0$, and therefore

$$f(x) = f(y) = f\left(\frac{x+y}{2}\right) = 1.$$

One property, several equivalent views

Let $S \subseteq \mathbb{R}^n$ be open and convex and let $f : S \rightarrow \mathbb{R}$ be twice continuously differentiable.

Equivalent statements for convexity. Each statement is equivalent to f being convex:

- ◆ f satisfies the chord inequality $f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$.
- ◆ $f(y) \geq f(x) + \nabla f(x)^\top (y - x)$ for all $x, y \in S$.
- ◆ $H_f(x) \succeq 0$ for every $x \in S$.

The list characterizes the weak property. A positive definite Hessian everywhere gives a one-way test for strict convexity.

One property, several equivalent views

Let $S \subseteq \mathbb{R}^n$ be open and convex and let $f : S \rightarrow \mathbb{R}$ be twice continuously differentiable.

Equivalent statements for concavity. Each statement is equivalent to f being concave:

- ◆ f satisfies the reversed chord inequality $f(\lambda x + (1 - \lambda)y) \geq \lambda f(x) + (1 - \lambda)f(y)$.
- ◆ $f(y) \leq f(x) + \nabla f(x)^\top (y - x)$ for all $x, y \in S$.
- ◆ $H_f(x) \preceq 0$ for every $x \in S$.

The list characterizes the weak property. A negative definite Hessian everywhere gives a one-way test for strict concavity.

Two further equivalent views

Epigraphs and subgradients in the same list

With S and f as on the preceding pages, convexity of f is also equivalent to each of

- ◆ $\text{epi} f = \{(x, t) : t \geq f(x)\}$ is a convex set;
- ◆ $\partial f(x) = \{g : f(y) \geq f(x) + g^\top (y - x) \ \forall y \in S\} \neq \emptyset$ for every $x \in S$.

Concavity of f is equivalent to each of

- ◆ $\text{hypo} f = \{(x, t) : t \leq f(x)\}$ is a convex set;
- ◆ $\partial^+ f(x) = \{g : f(y) \leq f(x) + g^\top (y - x) \ \forall y \in S\} \neq \emptyset$ for every $x \in S$.

Section 5

Quasiconvexity & quasiconcavity

Definitions & contour-set link

Let $S \subseteq \mathbb{R}^n$ be convex and $f : S \rightarrow \mathbb{R}$. For a level $c \in \mathbb{R}$,

$$U_f(c) = \{x \in S : f(x) \geq c\}, \quad L_f(c) = \{x \in S : f(x) \leq c\},$$

the upper and lower contour sets. The level set from the Functions section is their intersection, $\{x : f(x) = c\} = U_f(c) \cap L_f(c)$.

Quasiconvex. $f(\lambda x + (1 - \lambda)y) \leq \max\{f(x), f(y)\}$ for all $x, y \in S$ and $\lambda \in [0, 1]$; equivalently, every lower contour set $L_f(c)$ is convex.

Quasiconcave. $f(\lambda x + (1 - \lambda)y) \geq \min\{f(x), f(y)\}$ for all $x, y \in S$ and $\lambda \in [0, 1]$; equivalently, every upper contour set $U_f(c)$ is convex.

Videos: EZ Economics, *Understanding quasiconcave and quasiconvex functions*; Arizona Math Camp, *Upper and lower contour sets and quasiconcavity* (from 16:55).

Convexity versus quasiconvexity

f convex $\Rightarrow f$ quasiconvex $\Leftrightarrow L_f(c)$ convex for every c ,

f concave $\Rightarrow f$ quasiconcave $\Leftrightarrow U_f(c)$ convex for every c .

The first arrow runs one way; the second is an equivalence.

THE IMPLICATION IN ONE LINE

If f is convex, then

$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$,
and a weighted average is at most the maximum, so the right side is at most $\max\{f(x), f(y)\}$. Reversing the inequalities gives the concave case with min.

WHAT EACH SIDE ASKS OF THE GRAPH

Convexity constrains the whole graph: every chord lies above it, and every lower contour set is then convex. Quasiconvexity asks only that each horizontal slice $L_f(c)$ be convex, so it can hold where convexity fails.

Sign reversal and contour sets

Let $S \subseteq \mathbb{R}^n$ be convex and let $f : S \rightarrow \mathbb{R}$. Prove directly from contour sets that

$$f \text{ is quasiconvex} \iff -f \text{ is quasiconcave.}$$

State the symmetric result obtained by interchanging “quasiconvex” and “quasiconcave.”

Sign reversal and contour sets

For every $c \in \mathbb{R}$,

$$U_{-f}(c) = \{x : -f(x) \geq c\} = \{x : f(x) \leq -c\} = L_f(-c).$$

Therefore every lower contour set of f is convex if and only if every upper contour set of $-f$ is convex. Hence

$$f \text{ quasiconvex} \iff -f \text{ quasiconcave.}$$

Replacing f by $-f$ gives the mirror statement

$$f \text{ quasiconcave} \iff -f \text{ quasiconvex.}$$

This is the quasiconvex/quasiconcave analogue of f convex $\iff -f$ concave.

Quasiconcave but not concave

Let $u(x, y) = xy$ on $\mathbb{R}_{++}^2$. If $c \leq 0$, then $U_u(c) = \mathbb{R}_{++}^2$, which is convex. If $c > 0$,

$$U_u(c) = \{(x, y) : xy \geq c\} = \{(x, y) : y \geq c/x, x > 0\}.$$

Since $d^2(c/x)/dx^2 = 2c/x^3 > 0$ for $x > 0$, the function c/x is convex, so $g(x, y) = c/x - y$ is convex on $\mathbb{R}_{++} \times \mathbb{R}$ (a convex function of x plus a linear one). Thus

$U_u(c) = L_g(0) \cap (\mathbb{R}_{++} \times \mathbb{R})$ is a lower contour set of a convex function intersected with a convex set, hence convex.

Every upper contour set of u is convex, and therefore u is quasiconcave.

$$H_u = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

The Hessian is indefinite, so u is not concave.

Two-dimensional contour sets

On $S = \mathbb{R}_{++}^2$, classify each function as quasiconvex, quasiconcave, both, or neither:

$$u(x, y) = xy, \quad v(x, y) = \frac{x}{y}.$$

Use contour sets. For every negative or zero level, first identify whether the contour set at that level is empty or all of S ; for positive levels, write the contour set as an inequality in x and y .

Two-dimensional contour sets

For $u(x, y) = xy$, every upper contour set is convex, as shown on the preceding example page, so u is quasiconcave. Quasiconvexity fails: at level $c = 4$,

$$(1, 4), (4, 1) \in L_u(4), \quad \frac{1}{2}(1, 4) + \frac{1}{2}(4, 1) = \left(\frac{5}{2}, \frac{5}{2}\right),$$

but $(5/2)(5/2) = 25/4 > 4$, so the midpoint lies outside $L_u(4)$.

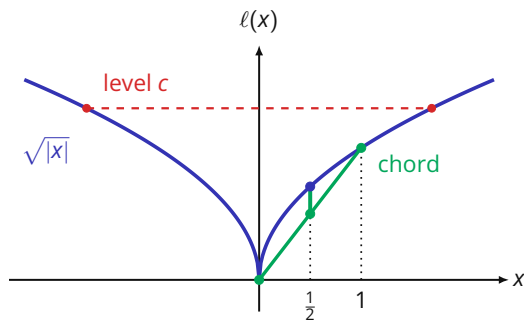
For $v(x, y) = x/y > 0$, if $c \leq 0$, then $L_v(c) = \emptyset$ and $U_v(c) = S$. If $c > 0$,

$$L_v(c) = \{(x, y) \in S : x \leq cy\}, \quad U_v(c) = \{(x, y) \in S : x \geq cy\}.$$

Each is the intersection of S with a half-space, hence convex. Therefore v is both quasiconvex and quasiconcave.

Quasiconvex but not convex

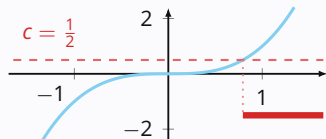
A forecaster incurs a loss based on the signed forecast error $x \in \mathbb{R}$: $\ell(x) = \sqrt{|x|}$. The loss increases with the absolute error, but at a decreasing rate. For $c \geq 0$, its lower contour sets are intervals, $\{x : \ell(x) \leq c\} = [-c^2, c^2]$; for $c < 0$, they are empty. Hence ℓ is quasiconvex. The chord joining $(0, \ell(0))$ and $(1, \ell(1))$ settles the other half: at its midpoint $\ell(1/2) = 1/\sqrt{2} > 1/2 = [\ell(0) + \ell(1)]/2$, so the graph rises above the chord and convexity fails.



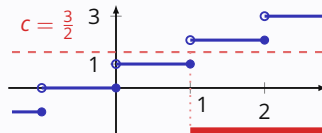
Red level line: the lower contour set is an interval. Green chord: at $x = \frac{1}{2}$ the graph sits $\frac{1}{\sqrt{2}} - \frac{1}{2}$ above it (the short vertical segment).

Three functions that test the definition

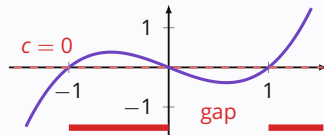
Figure 5 · The bar under each axis is the upper contour set at the level drawn.



$f(x) = x^3$. Each upper contour set is a half-line, so f is quasiconcave, and so is $-f$. Yet $f''(x) = 6x$ changes sign, so f is neither concave nor convex.



$f(x) = \lceil x \rceil$. The definition asks nothing about continuity: this step function is quasiconcave and quasiconvex while jumping at every integer.



$f(x) = x^3 - x$. Now $U_f(0) = [-1, 0] \cup [1, \infty)$ falls into two pieces, so f is not quasiconcave, though it adds the quasiconcave x^3 to the linear $-x$.

The step function is Example 3.29 of [Boyd and Vandenberghe](#). The right panel is where quasiconcavity parts company with concavity: a sum of concave functions stays concave, a sum of quasiconcave functions need not stay quasiconcave.

More examples: [EconJohn, Quasiconcave and quasiconvex functions: an introduction](#).

Quasiconvexity and quasiconcavity

For each function below, determine whether it is quasiconvex, quasiconcave, both, or neither on $\mathbb{R}$:

$$f(x) = e^x, \quad g(x) = x^2, \quad h(x) = -|x|.$$

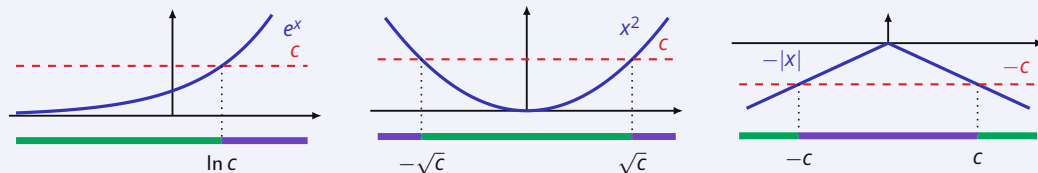
Justify every answer using upper or lower contour sets. Do not use a Hessian test.

Quasiconvexity and quasiconcavity

e^x is strictly increasing. Its upper and lower contour sets are half-lines, so it is both quasiconvex and quasiconcave.

The lower contour sets of x^2 are intervals $[-\sqrt{c}, \sqrt{c}]$ for $c \geq 0$, so x^2 is quasiconvex. For $c > 0$, its upper contour set is the union of two rays, so it is not quasiconcave.

The upper contour sets of $-|x|$ are intervals $[-c, c]$ after writing the nonempty levels as $c \geq 0$ for $-|x| \geq -c$. Hence $-|x|$ is quasiconcave. Its lower contour sets can be two rays, so it is not quasiconvex.



Below each graph, at the dashed level: green marks the lower contour set L , purple the upper contour set U . An interval or half-line is convex; two rays are not.

Convex preferences

Let $u : S \rightarrow \mathbb{R}$ represent a preference relation $\succeq$ on a convex set S . Preferences are **convex** when

$$x \succeq z, y \succeq z \Rightarrow \lambda x + (1 - \lambda)y \succeq z \quad \forall \lambda \in [0, 1].$$

Since $\{x : x \succeq z\} = \{x : u(x) \geq u(z)\}$,

$$\boxed{\succeq \text{ is convex} \Leftrightarrow u \text{ is quasiconcave.}}$$

Strict quasiconcavity requires, for $x \neq y$ and $\lambda \in (0, 1)$,

$$u(\lambda x + (1 - \lambda)y) > \min\{u(x), u(y)\}.$$

Under the corresponding strict-convexity definition for preferences, the strict versions are equivalent as well.

Preference convexity is a property of the upper contour sets. Concavity of a utility representation is stronger: it implies quasiconcavity, while an increasing transformation can change concavity.

Video: [Jochumzen, Convex preferences.](#)

Convex preferences and strict quasiconcavity

On $S = \mathbb{R}_+^2$, consider the perfect-complements utility function

$$u(x_1, x_2) = \min\{x_1, 2x_2\}.$$

- (a) Describe $U_u(c)$ and use it to show that u is quasiconcave.
- (b) What does part (a) imply about the preferences represented by u ?
- (c) Is u strictly quasiconcave? Give a pair of distinct bundles that settles the question.

Convex preferences and strict quasiconcavity

For $c \leq 0$, $U_u(c) = S$. For $c > 0$, $\min\{x_1, 2x_2\} \geq c$ holds iff $x_1 \geq c$ and $x_2 \geq c/2$, so

$$U_u(c) = S \cap \{x_1 \geq c\} \cap \{x_2 \geq c/2\},$$

an intersection of convex sets, hence convex. Thus u is quasiconcave, and the preferences it represents are convex.

Strict quasiconcavity fails. Take $x = (2, 1)$ and $y = (4, 1)$. Then $u(x) = u(y) = 2$, while for every $\lambda \in (0, 1)$,

$$u(\lambda x + (1 - \lambda)y) = u(4 - 2\lambda, 1) = 2 = \min\{u(x), u(y)\}.$$

The required strict inequality therefore fails along the whole segment.

Ordinal transformations

Compare, on $\mathbb{R}_{++}^2$,

$$u(x, y) = xy, \quad v(x, y) = \ln x + \ln y = \ln u(x, y).$$

Since $\ln$ is strictly increasing,

$$u(x, y) \geq u(x', y') \Leftrightarrow v(x, y) \geq v(x', y').$$

Thus u and v represent the same preferences.

Here $H_v = \text{diag}(-1/x^2, -1/y^2)$ is negative definite, so v is concave, while $H_u = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ has eigenvalues ± 1 , so u is indefinite there.

Here $u = \exp(v)$ with $\exp$ increasing: an increasing transformation of a concave function need not be concave. Quasiconcavity does survive.

Quick check. Let u be quasiconcave and let ϕ be strictly increasing. Show that $\phi \circ u$ is quasiconcave. Give an example showing that concavity need not be preserved.

Increasing transformations

For $z = tx + (1 - t)y$, quasiconcavity gives

$$u(z) \geq \min\{u(x), u(y)\}.$$

Since ϕ is strictly increasing,

$$\phi(u(z)) \geq \phi(\min\{u(x), u(y)\}) = \min\{\phi(u(x)), \phi(u(y))\}.$$

Hence $\phi \circ u$ is quasiconcave. For the second claim, take $u(x) = x$ and $\phi(t) = e^t$ on $\mathbb{R}$. The function u is concave, while $(\phi \circ u)(x) = e^x$ is not concave.

First-order characterization

Let $S \subseteq \mathbb{R}^n$ be **open and convex** and let $f : S \rightarrow \mathbb{R}$ be differentiable.

Quasiconcave iff.

$$f(y) \geq f(x) \implies \nabla f(x)^\top (y - x) \geq 0 \quad \text{for all } x, y \in S.$$

At a point x with $\nabla f(x) \neq 0$, the upper contour set $U_f(f(x))$ lies in $\{y : \nabla f(x)^\top (y - x) \geq 0\}$, the side of the tangent hyperplane to the level set into which the gradient points. The concave case in the first-order section bounds the value itself, $f(y) \leq f(x) + \nabla f(x)^\top (y - x)$.

Quasiconcave $\Rightarrow$ inequality. Fix x, y with $f(y) \geq f(x)$ and set $\varphi(t) = f(x + t(y - x))$ on $[0, 1]$. Quasiconcavity gives $\varphi(t) \geq \min\{f(x), f(y)\} = \varphi(0)$, so $\varphi'(0) = \nabla f(x)^\top (y - x) \geq 0$.

Source: Sydsæter et al., *Further Mathematics for Economic Analysis*, 2nd ed., Theorem 2.5.4, which proves this direction; the converse is on the next page.

First-order characterization: the converse

Inequality $\Rightarrow$ quasiconcave. Suppose the implication holds for all pairs, and take $x, y \in S$ with $f(y) \geq f(x)$. We show that $\varphi(t) = f(x + t(y - x)) \geq f(x)$ on $[0, 1]$.

If $\varphi(\bar{t}) < f(x)$ for some $\bar{t} \in (0, 1)$, let $t_1 = \max\{t \in [0, \bar{t}] : \varphi(t) \geq f(x)\}$, which exists by continuity; then $\varphi(t_1) = f(x)$ and $\varphi(t) < f(x)$ on $(t_1, \bar{t}]$. For such t , write $z = x + t(y - x)$:

- ◆ $f(x) > f(z)$ gives $\nabla f(z)^\top (x - z) \geq 0$, and $x - z = -t(y - x)$, so $\varphi'(t) = \nabla f(z)^\top (y - x) \leq 0$;
- ◆ $f(y) \geq f(x) > f(z)$ gives $\nabla f(z)^\top (y - z) \geq 0$, and $y - z = (1 - t)(y - x)$, so $\varphi'(t) \geq 0$.

Hence $\varphi' = 0$ on $(t_1, \bar{t}]$, so φ is constant there, equal to $\varphi(\bar{t}) < f(x)$. Continuity at t_1 then forces $\varphi(t_1) = \varphi(\bar{t}) < f(x)$, contradicting $\varphi(t_1) = f(x)$. Therefore $\varphi \geq f(x) = \min\{f(x), f(y)\}$ on $[0, 1]$, which is quasiconcavity.

First-order tests for quasiconcavity and quasiconvexity

- (a) Apply the first-order characterization to $-f$ and derive the corresponding differentiable test for quasiconvexity.
- (b) Using only the first-order test, show that $f(x_1, x_2) = x_1x_2$ is quasiconcave on $\mathbb{R}_{++}^2$.

First-order tests for quasiconcavity and quasiconvexity

(a) Since f is quasiconvex iff $-f$ is quasiconcave, apply the quasiconcavity criterion to $-f$:

$$-f(y) \geq -f(x) \implies \nabla[-f](x)^\top (y - x) \geq 0.$$

Multiplying the antecedent and the consequent by -1 gives

$$f(y) \leq f(x) \implies \nabla f(x)^\top (y - x) \leq 0.$$

(b) Write $x = (a, b)$ and $y = (c, d)$, with all four coordinates positive, and suppose $f(y) \geq f(x)$, so $cd \geq ab$. Since $\nabla f(a, b) = (b, a)$,

$$\nabla f(x)^\top (y - x) = b(c - a) + a(d - b) = ab \left(\frac{c}{a} + \frac{d}{b} - 2 \right).$$

Put $r = c/a$ and $s = d/b$, both positive, with $rs = cd/(ab) \geq 1$. The inequality between weighted arithmetic and geometric means (the exercise in the second-order section, with weights $\frac{1}{2}$) gives $r + s \geq 2\sqrt{rs} \geq 2$. Hence $\nabla f(x)^\top (y - x) \geq 0$, and the first-order characterization proves that $x_1 x_2$ is quasiconcave on $\mathbb{R}_{++}^2$.

Bordered Hessian test in two variables

A determinant criterion for quasiconcavity

Let $S \subseteq \mathbb{R}^2$ be **open and convex** and $f : S \rightarrow \mathbb{R}$ twice continuously differentiable. Border the Hessian with the gradient:

$$B_2(x) = \det \begin{bmatrix} 0 & f_1 & f_2 \\ f_1 & f_{11} & f_{12} \\ f_2 & f_{21} & f_{22} \end{bmatrix} = 2f_1f_2f_{12} - f_1^2f_{22} - f_2^2f_{11}.$$

Necessary. If f is quasiconcave on S , then $B_2(x) \geq 0$ for every $x \in S$.

Sufficient. If $B_2(x) > 0$ for every $x \in S$, then f is strictly quasiconcave on S .

Source: Sydsæter et al., *Further Mathematics for Economic Analysis*, 2nd ed., Theorem 2.5.5; their sufficient condition also assumes $f_1 \neq 0$, which the proof on the next pages does without.

Bordered Hessian test: proof

Restrict to a line tangent to the level curve

Both parts use $\varphi(t) = f(p + tv)$ for a point $p \in S$ and a direction $v \neq 0$ with $\nabla f(p)^\top v = 0$; then $\varphi'(0) = 0$ and $\varphi''(0) = v^\top H_f(p)v$. In the plane such a v is a multiple of $(-f_2(p), f_1(p))$ whenever $\nabla f(p) \neq 0$, so $\varphi''(0) = -c^2 B_2(p)$ for some $c \neq 0$.

Necessary. If $\nabla f(p) = 0$, then $B_2(p) = 0$. Otherwise suppose $B_2(p) < 0$, so $\varphi''(0) > 0$. By Taylor's theorem $\varphi(t) > \varphi(0)$ for all small $t \neq 0$: both $p + tv$ and $p - tv$ have values above $f(p)$, while quasiconcavity at their midpoint p gives $f(p) \geq \min\{f(p + tv), f(p - tv)\} > f(p)$, a contradiction. Hence $B_2(p) \geq 0$.

Bordered Hessian test: proof of sufficiency

A minimum on the segment would be a tangential minimum

Sufficient. Suppose $B_2 > 0$ on S but f is not strictly quasiconcave: there are $x \neq y$ and $\lambda \in (0, 1)$ with $f(\lambda x + (1 - \lambda)y) \leq \min\{f(x), f(y)\}$. Then $\psi(t) = f(x + t(y - x))$ has a value at λ that is at most both endpoint values, so it attains its minimum over $[0, 1]$ at some $t_0 \in (0, 1)$, where $\psi'(t_0) = 0$ and $\psi''(t_0) \geq 0$. Put $p = x + t_0(y - x)$ and $v = y - x \neq 0$. Since B_2 vanishes wherever the gradient does, $B_2(p) > 0$ gives $\nabla f(p) \neq 0$, and $\psi'(t_0) = \nabla f(p)^\top v = 0$ makes v tangent to the level curve, so $\psi''(t_0) = v^\top H_f(p)v = -c^2 B_2(p) < 0$, contradicting $\psi''(t_0) \geq 0$. Hence f is strictly quasiconcave.

Cobb–Douglas utility

Strictly quasiconcave for every pair of positive exponents

Let $u(x_1, x_2) = x_1^\alpha x_2^\beta$ on $\mathbb{R}_{++}^2$ with $\alpha, \beta > 0$. Then

$$u_1 = \frac{\alpha u}{x_1}, \quad u_2 = \frac{\beta u}{x_2}, \quad u_{11} = \frac{\alpha(\alpha - 1)u}{x_1^2}, \quad u_{22} = \frac{\beta(\beta - 1)u}{x_2^2}, \quad u_{12} = \frac{\alpha\beta u}{x_1 x_2}.$$

Substituting into $B_2 = 2u_1 u_2 u_{12} - u_1^2 u_{22} - u_2^2 u_{11}$ and collecting $u^3/(x_1 x_2)^2$,

$$B_2 = \frac{u^3}{(x_1 x_2)^2} [2\alpha^2 \beta^2 - \alpha^2 \beta(\beta - 1) - \alpha \beta^2(\alpha - 1)] = \frac{\alpha \beta (\alpha + \beta) u^3}{(x_1 x_2)^2} > 0,$$

so u is strictly quasiconcave for all $\alpha, \beta > 0$. Concavity is a different matter:

$\det H_u = u_{11} u_{22} - u_{12}^2 = \alpha \beta (1 - \alpha - \beta) u^2 / (x_1 x_2)^2$ is negative when $\alpha + \beta > 1$; the Hessian is then indefinite (the case $\alpha = \beta = 1$ is on the ordinal transformations page), so u is quasiconcave but neither concave nor convex. Source: Sydsæter et al., *Further Mathematics for Economic Analysis*, 2nd ed., §2.5, the example after Theorem 2.5.5.

Bordered Hessian versus the ordinary Hessian

Strict quasiconcavity without concavity

On $S = \mathbb{R}_{++}^2$, let

$$f(x_1, x_2) = x_1 + x_2 + x_1 x_2.$$

- (a) Compute the ordinary Hessian and decide whether f is convex or concave.
- (b) Compute the bordered determinant B_2 and use it to classify the quasiconcavity of f .
- (c) Explain in one sentence why the two answers are compatible.

Bordered Hessian versus the ordinary Hessian

Different curvature requirements

The derivatives are

$$f_1 = 1 + x_2, \quad f_2 = 1 + x_1, \quad H_f = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

The Hessian has eigenvalues 1 and -1 , so it is indefinite everywhere; hence f is neither convex nor concave on the open convex set S .

For the bordered determinant,

$$B_2 = 2f_1f_2f_{12} - f_1^2f_{22} - f_2^2f_{11} = 2(1 + x_2)(1 + x_1) > 0$$

everywhere on S . The sufficient bordered-Hessian test therefore makes f strictly quasiconcave.

The two answers are compatible: concavity controls the curvature of f in every direction, whereas quasiconcavity requires convex upper contour sets and only restricts curvature in directions tangent to a level set.

Bordered Hessians in n variables

Alternating signs of the leading bordered minors

Let $S \subseteq \mathbb{R}^n$ be open and convex and $f : S \rightarrow \mathbb{R}$ twice continuously differentiable. For $r = 1, \dots, n$ let

$$B_r(x) = \det \begin{bmatrix} 0 & f_1 & \cdots & f_r \\ f_1 & f_{11} & \cdots & f_{1r} \\ \vdots & \vdots & & \vdots \\ f_r & f_{r1} & \cdots & f_{rr} \end{bmatrix}.$$

- ◆ If f is quasiconcave on S , then $(-1)^r B_r(x) \geq 0$ for all $x \in S$ and $r = 1, \dots, n$.
- ◆ If $(-1)^r B_r(x) > 0$ for all $x \in S$ and $r = 1, \dots, n$, then f is strictly quasiconcave on S .

For $r = 1$, $(-1)B_1 = f_1^2$, so the strict condition requires $f_1 \neq 0$ on S ; for $n = 2$ the condition with $r = 2$ is $B_2 > 0$, the test of the previous pages.

Sources: Sydsæter et al., *Further Mathematics for Economic Analysis*, 2nd ed., Thm 2.5.6; Arrow and Enthoven (1961), *Econometrica* 29, 779–800.